

Numark

V7

(NK14)

Service Manual

Rev. 1
2010.04.13

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PREFACE

This document is intended to assist the service technician in the operation, maintenance and repair of the Numark device. Together with the User Reference Manual, this document provides a complete description of the functionality and serviceability of the Device. Any comments or suggestions you may have pertaining to the document are welcome and encouraged.

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WARNINGS

TO REDUCE THE RISK OF ELECTRIC SHOCK OR FIRE, DO NOT EXPOSE THIS PRODUCT TO WATER OR MOISTURE.



The arrowhead symbol on a lightning flash inside a triangle is intended to alert the user to the presence of uninsulated "dangerous voltage" within the enclosed product which may be of sufficient magnitude to constitute a risk of electric shock to persons.



The exclamation point inside a triangle is intended to alert the user to the presence of important operating, maintenance and servicing instructions in the literature which accompanies the product.

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Regarding the Power Supply Fuse



CAUTION: The product under service may employ the use of a replaceable fuse. Danger of fire or electrocution if fuse is incorrectly replaced. Replace with only the same type or equivalent type recommended by the equipment manufacturer.

Regarding the Internal Battery



CAUTION: The product under service may employ the use of a internal battery. Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Dispose of used batteries according to the manufacturer's instruction.

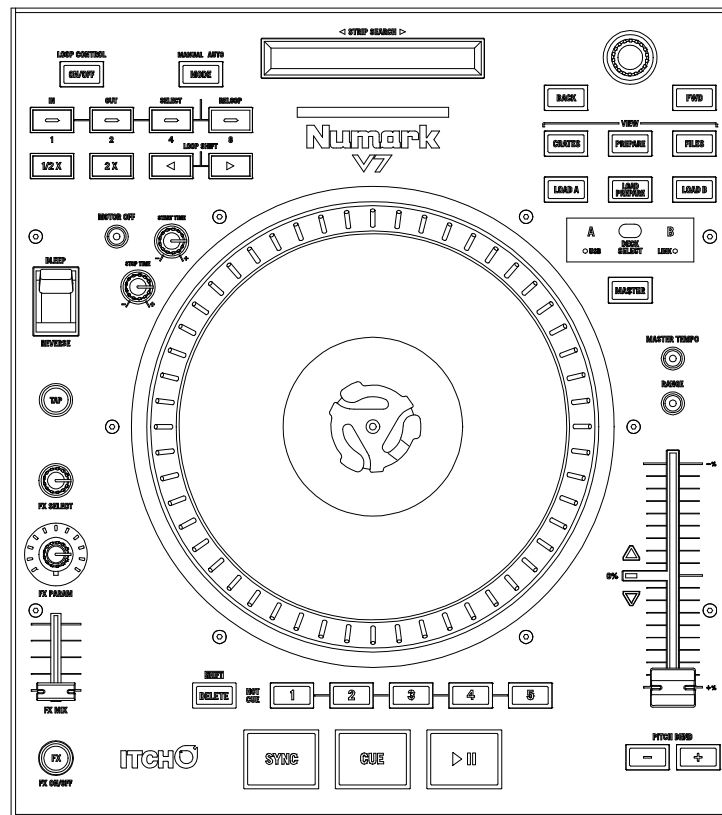
Safety Instructions

Carefully read the applicable items of the operating instructions and these safety suggestions before using this product. Use extra care to follow the warnings written on the product itself and in the operating instructions. Keep the operating instructions and safety suggestions for reference in the future.

1. Power Source. The product should only be connected to a power supply which is described either in the operating instructions or in markings on the product.
2. Power Cord Protection. AC power supply cords should be placed such that no one is likely to step on the cords and such that nothing will be placed on or against them.
3. Periods of Non-use. If the product is not used for any significant period of time, the product's AC power supply cord should be unplugged from the AC outlet.
4. Foreign Objects and Liquids. Take care not to allow liquids to spill or objects to fall into any openings of the product.
5. Water or Moisture. The product should not be used near any water or in moisture.
6. Heat. Do not place the product near heat sources such as stoves, heat registers, radiators or other heat producing equipment.
7. Ventilation. When installing the product, make sure that the product has adequate ventilation. Improperly ventilating the product may cause overheating, which may damage the product.
8. Mounting. The product should only be used with a rack which the manufacturer recommends. The combination of the product and rack should be moved carefully. Quick movements, excessive force or uneven surfaces may overturn the combination which may damage the product and rack combination.
9. Cleaning. The product should only be cleaned as the manufacturer recommends.
10. Service. The user should only attempt the limited service or upkeep specifically described in the operating instructions for the user. For any other service required, the product should be taken to an authorized service center as described in the operating instructions.
11. Damage to the Product. Qualified service personnel should service the unit in certain situations including without limitation when:
 - a. Liquid has spilled or objects have fallen into the product,
 - b. The product is exposed to water or excessive moisture,
 - c. The AC power supply plug or cord is damaged,
 - d. The product shows an inappropriate change in performance or does not operate normally, or
 - e. The enclosure of the product has been damaged.

SERVICE MANUAL

MODEL:V7



SPECIFICATION

FUNCTION

1. **MOTOR TORQUE** – Flip this switch to adjust the torque of the PLATTERS. At the high setting, the PLATTERS will have the heavier, stronger feel of “modern” turntables. At the lower setting, they are lighter and more graceful – the feel of a “classic” turntable.
2. **DECK A / DECK B OUTPUT (RCA)** – Use standard RCA cables to connect these outputs for Deck A and Deck B to the left and right channels (respectively) of your mixer.
3. **DECK LOCATION SWITCH** – Reserved for future use.
4. **LINK CONNECTION** – If you are using two V7s, connect their LINK CONNECTIONS with the included link cable.
5. **START TIME / STOP TIME** – Control the rate at which the playback shifts from “play” to “pause” and from “pause” to “play.”
6. **BLEEP / REVERSE** – Reverses audio playback of the track on the corresponding deck.
7. **SYNC** – Automatically matches the corresponding Deck’s tempo with the other Deck’s tempo.
8. **IN** – Press this to set a “Loop In” point where your loop will begin. If you are already in a loop when you press it, you can “fine tune” the placement of the “Loop In” point by moving the PLATTER.
9. **OUT** – Press this to set a “Loop Out” point where your loop will end. If you are already in a loop when you press it, you can “fine tune” the placement of the “Loop Out” point by moving the PLATTER.
10. **RELOOP** – Press this to jump to the start of the loop and enable it. If the Deck is already looping when you press it, Pointer will jump to the start of the loop and continue. If no loop is set, nothing will happen.
11. **1/2 X** – Press this to halve the length of the selected loop.
12. **2 X** – Press this to double the length of the selected loop.
13. **STRIP SEARCH** – The length of this strip represents the length of the entire track. Place your finger on a point along this sensor to jump to that point in the track. (If you want to scroll through a track, we recommend using your computer rather than running your finger along the strip.)
14. **TAP** – Tap this button at the same tempo as the track to help the software detect a more accurate BPM reading.
15. **CRATES** – Press this to move the selector to the Crates Panel in the software.

STANDARD

1. MaxMSP test program for production Line
2. AP fest specification

RCA Output

Voltage	AC 100V~240V 50~60Hz
Volume	$2 \pm 0.5V$
Distortion defection	0.01% below
S/ N	96dBr upward
Separation	96dBr upward
F/R	20~20KHz $\pm 0.5dB$

DISASSEMBLY PROCEDURES

1. REMOVAL OF CABINET AND BOTTOM CHASSIS. (Fig1)

(A) REMOVAL 27 PCS OF SCREWS BEFORE REMOVE THE BOTTOM CHASSIS.

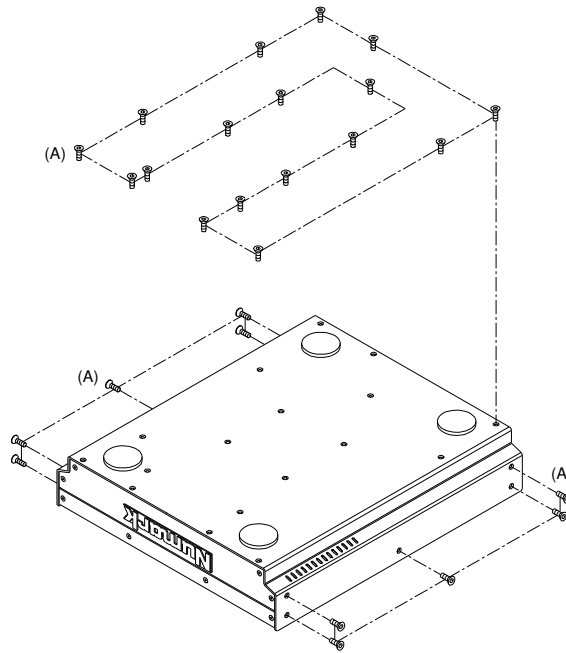


Fig.1

2.REMOVAL OF MOTOR CONTRON PCB AND MAIN PCB.(Fig.2)

(A) REMOVE THE 3 PCS CONNECTORS.

(B) REMOVE THE 4 PCS OF SCREWS BEFORE REMOVAL OF MOTOR CONTRON PCB.

(C) REMOVE THE 4 PCS SCREW.

(D) REMOVE THE 3 PCS OF CONNECTORS BEFORE REMOVAL OF MAIN PCB.

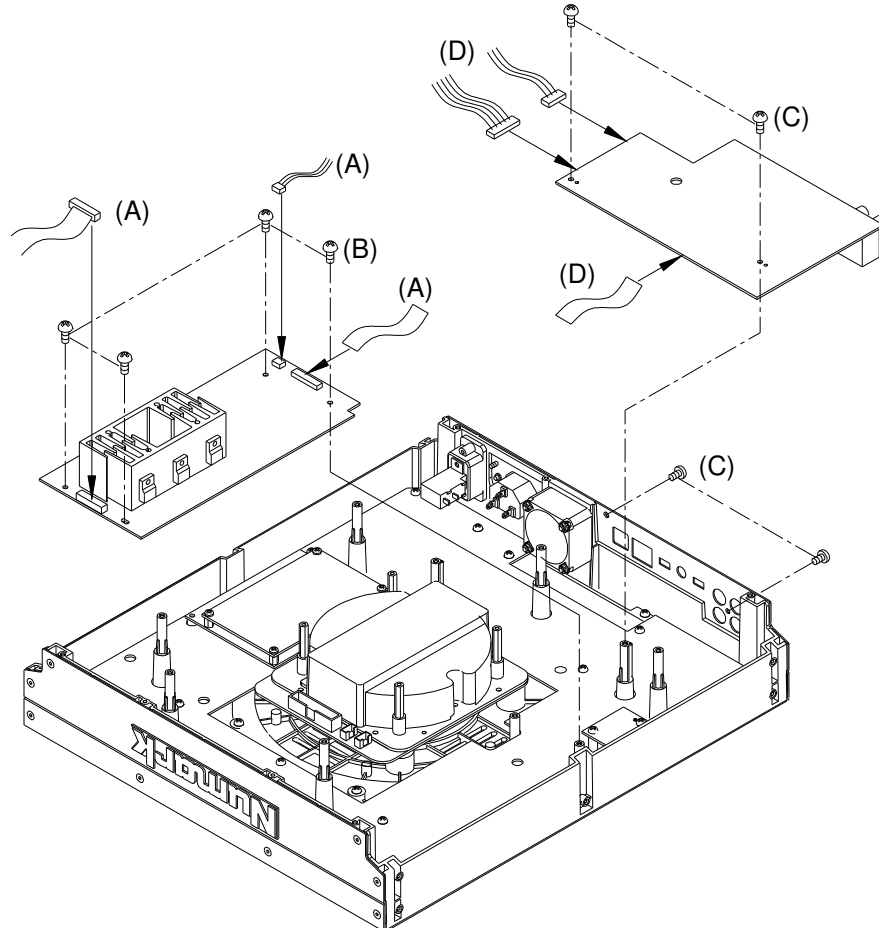


Fig.2

3.REMOVAL OF SWITCHING POWER AND Bleep/Reverse PCB .(Fig3)

- (A) REMOVE THE 1 PC OF CONNECTOR.
- (B) REMOVE THE 4 PCS SCREW.
- (C) REMOVE THE 3 PCS COPPER.
- (D) REMOVE THE 1 PC SCREW BEFORE REMOVAL OF SWITCHING PCB.
- (E) REMOVE THE 3 PCS OF SCREWS BEFORE REMOVAL OF Bleep/Reverse PCB.

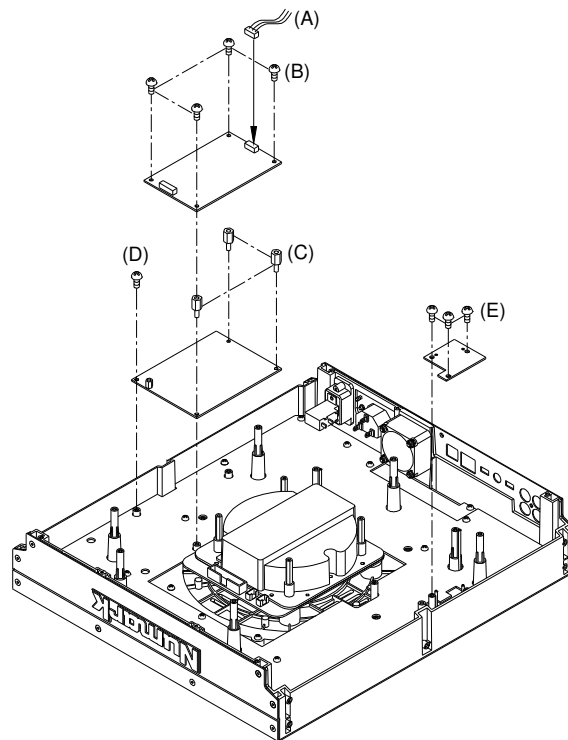


Fig.3

4.REMOVAL OF FUNTION PCB.(Fig4)

- (A) REMOVE THE 7 PCS OF VR KNOB.
- (B) REMOVE THE 4 PCS OF SCREW.
- (C) REMOVE THE 1 PC OF CONNECTOR.
- (D) REMOVE THE 22 PCS CONNECTOR BEFORE REMOVAL OF FUNTION PCB.

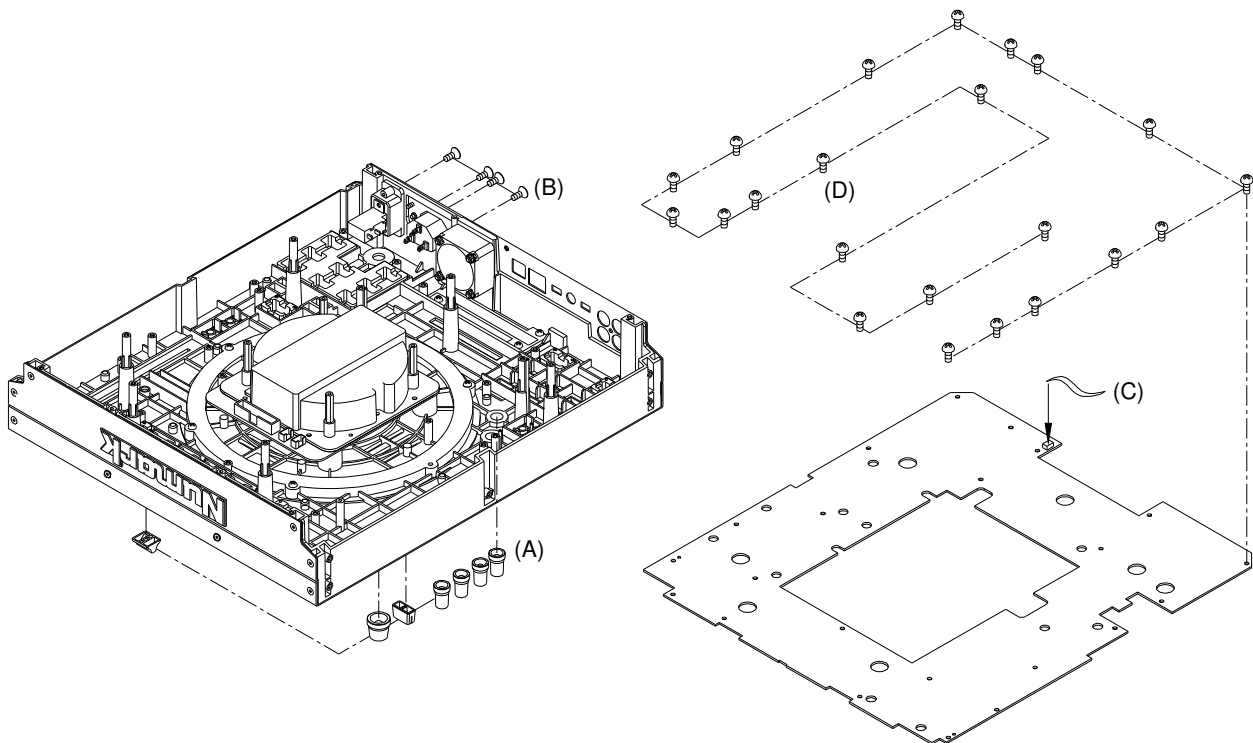
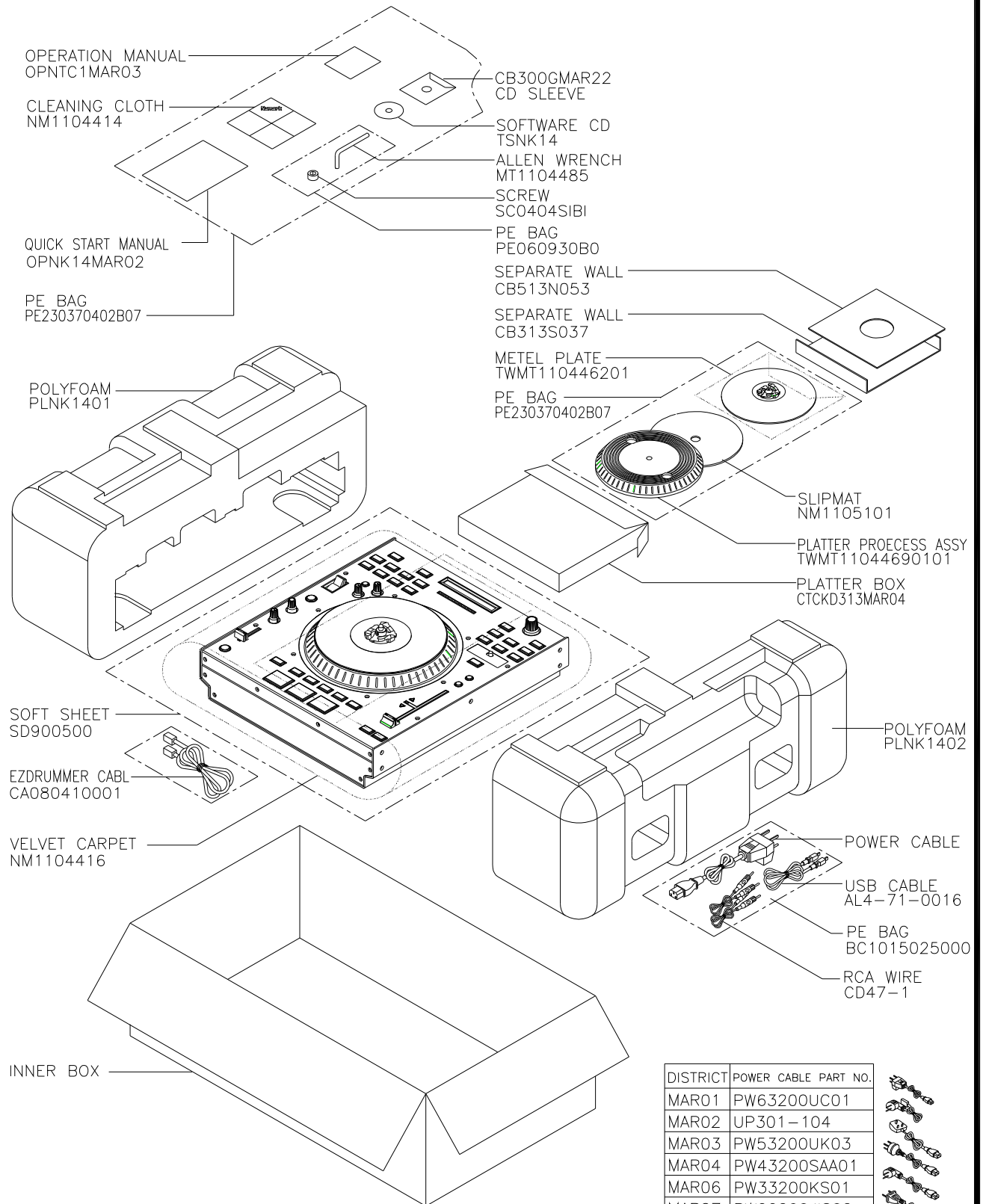


Fig.4

PACKING DIAGRAM

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NK14MAR01/02/03/04/06/07

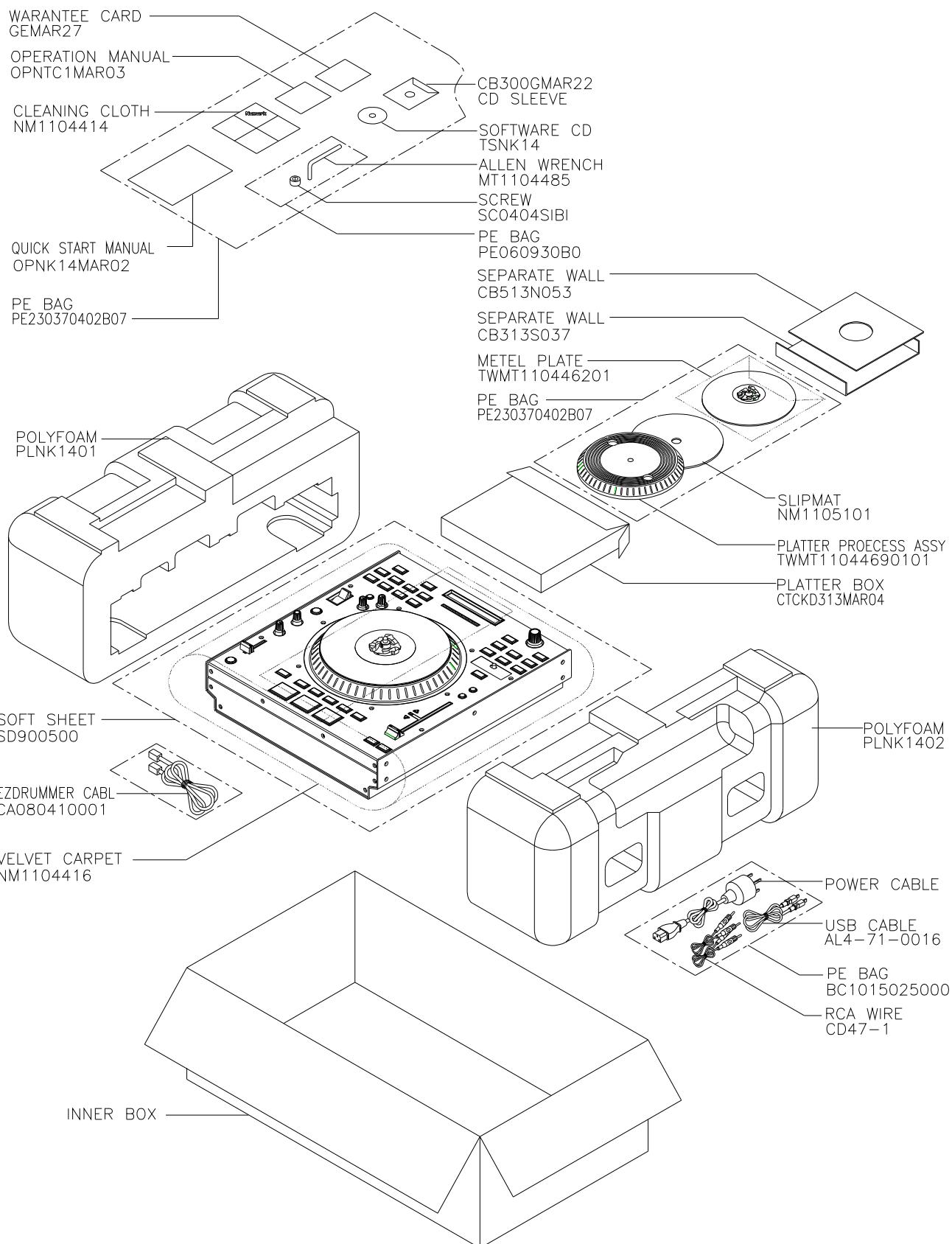


| DISTRICT | POWER CABLE PART NO. |
|----------|----------------------|
| MAR01    | PW63200UC01          |
| MAR02    | UP301-104            |
| MAR03    | PW53200UK03          |
| MAR04    | PW43200SAA01         |
| MAR06    | PW33200KS01          |
| MAR07    | PW22200JIS02         |

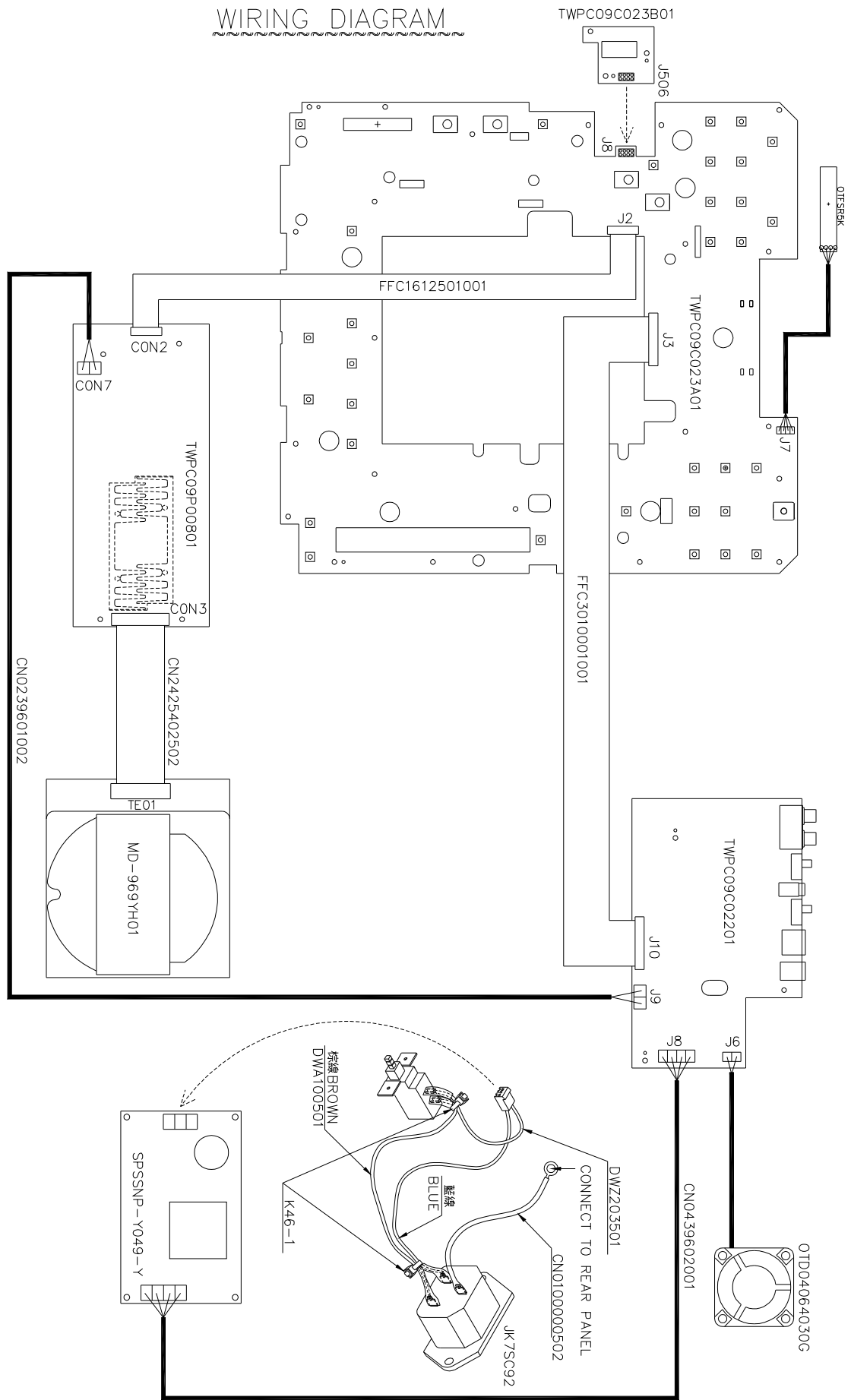


# PACKING DIAGRAM

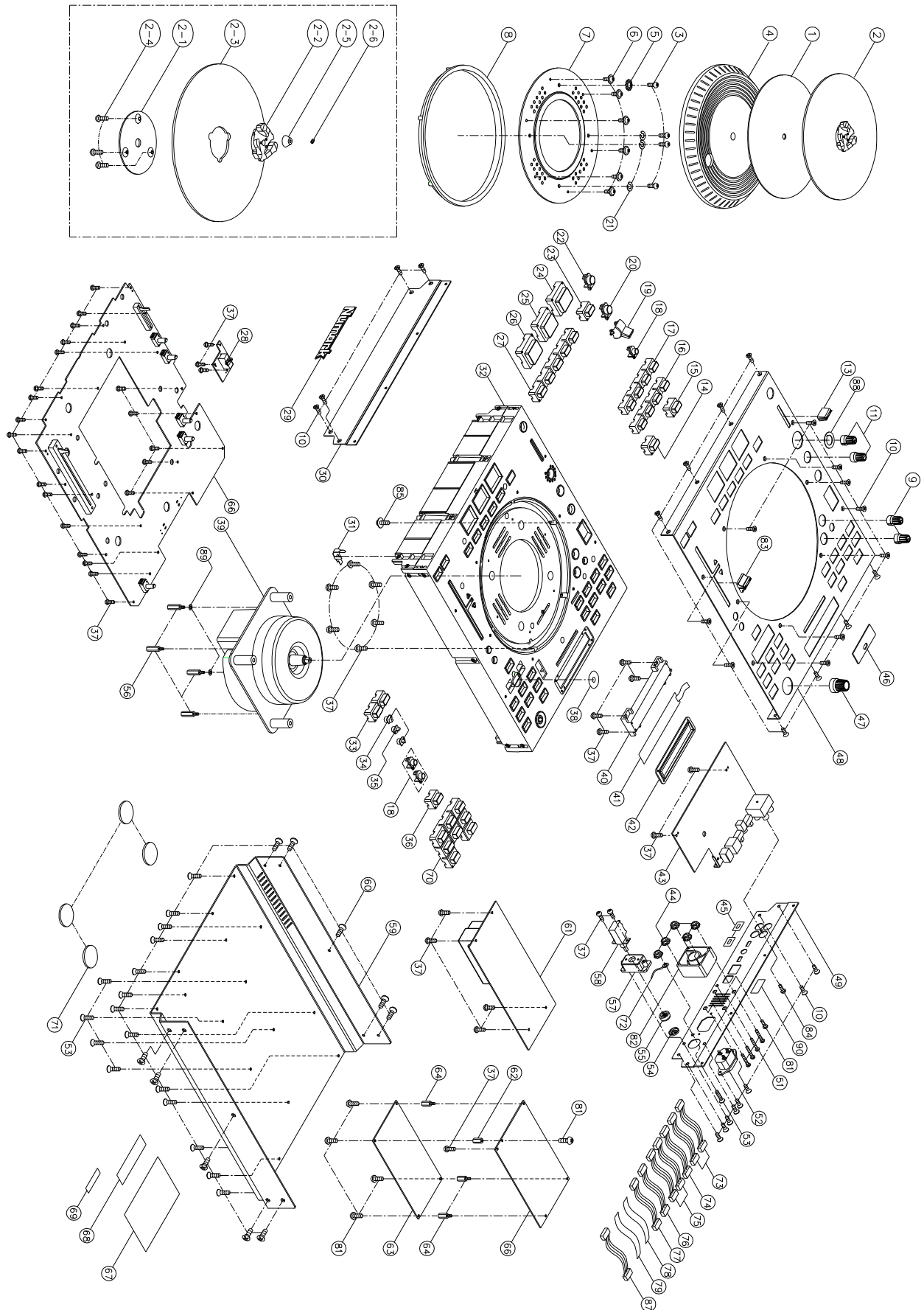
NK14MAR05



# WIRING DIAGRAM



# EXPLODE DIAGRAM



# NK14MAR01 BOM

| LEVEL | P/N           | DESCRIPTION                    | QTY | REF |
|-------|---------------|--------------------------------|-----|-----|
| 0.1   | BAGL-55A      | CABLE CLIP 55L                 | 2   |     |
| 0.1   | BAGL-70L      | CLIP                           | 2   |     |
| 0.1   | CA080410001   | Ethernet Cable                 | 1   |     |
| 0.1   | CB300GMAR22   | CD SLEEVE                      | 1   |     |
| 0.1   | CD47-1        | PHONO CORD                     | 2   |     |
| 0.1   | CN0100000502  | 1P CONNECTOR (50mm)            | 1   | 72  |
| 0.1   | CN0100000705  | 1P CONNECTOR (70mm)            | 2   | 73  |
| 0.1   | CN0100001605  | 1P CONNECTOR (160mm)           | 1   | 74  |
| 0.1   | CN0100000502  | 1P CONNECTOR (50mm)            | 2   | 75  |
| 0.1   | CN0239601002  | 2P CONNECTOR (100mm)           | 1   | 76  |
| 0.1   | CN0439602501  | 4P CONNECTOR (250mm)           | 1   | 87  |
| 0.1   | CN2425402502  | 24P CONNECTOR (250mm)          | 1   | 77  |
| 0.1   | FFC1612501001 | 16P CONNECTOR (100mm)          | 1   | 78  |
| 0.1   | FFC3010001001 | 30P CONNECTOR (100mm)          | 1   | 79  |
| 0.1   | JK7SC92       | AC SOCKET                      | 1   | 52  |
| 0.1   | K46-1         | CABLE TIE                      | 3   |     |
| 0.1   | LA62GMN37     | DATE LABEL                     | 1   | 69  |
| 0.1   | LA67YH104     | GND INDICATE STICKER           | 1   |     |
| 0.1   | LAC22MAR336   | S/N BAR CODE LABEL             | 1   | 68  |
| 0.1   | LAC56MAR381   | WARNING LABEL                  | 2   |     |
| 0.1   | LAC62MAR337   | BAR CODE LABEL                 | 2   |     |
| 0.1   | LAC63MAR338   | SEFETY LABEL                   | 1   | 67  |
| 0.1   | LAC66MAR382   | USB LABEL                      | 1   | 90  |
| 0.1   | LAC67YAH261   | STICKER φ8MM GREEN             | 1   |     |
| 0.1   | MD-969YH01    | DD MOTOR                       | 1   | 39  |
| 0.1   | MT1101818     | METAL PLATE                    | 1   | 31  |
| 0.1   | MT1105522     | COPPER                         | 3   | 64  |
| 0.1   | MT1105525     | BRASS STANDOFF                 | 4   | 56  |
| 0.1   | MT1105529     | COPPER                         | 1   | 62  |
| 0.1   | MT112022201   | CABINET PRINTING               | 1   | 7   |
| 0.1   | MT112045101   | FRONT PANEL PAINTING           | 1   | 30  |
| 0.1   | MT112045202   | REAR PANEL PRINTING            | 1   | 49  |
| 0.1   | MT113006101   | FRONT PANEL (ANDOIZED PRINTING | 1   | 48  |
| 0.1   | MT113033901   | BOTTOM CHASSIS PAINTING & PRIN | 1   | 59  |
| 0.1   | MT1310601     | POWER BUTTON CAP               | 1   | 54  |
| 0.1   | NM0704404     | VELVET CARPET                  | 2   | 45  |
| 0.1   | NM1104412     | VELVET CARPET                  | 1   | 38  |
| 0.1   | NP211MAR109   | NAME PLATE                     | 1   | 88  |
| 0.1   | NP211MAR99    | NAME PLATE(LEFT/RIGHT)         | 1   | 46  |
| 0.1   | NP311MAR92    | LOGO                           | 1   | 29  |
| 0.1   | NUT01030IID   | NUT                            | 5   | 44  |
| 0.1   | OTD04064030G  | DC FAN 40*40*20MM 12V          | 1   | 82  |
| 0.1   | OTFSR5K       | Force Sensing Resistor         | 1   | 41  |
| 0.1   | PT0710620801  | KNOB                           | 1   | 47  |
| 0.1   | PT0710621101  | KNOB                           | 2   | 9   |
| 0.1   | PT1104412601  | FIXED HOLDER(FSR)PAINTING      | 1   | 42  |
| 0.1   | PT1104412701  | FIXED HOLDER(TABLE)PAINTING    | 1   | 8   |
| 0.1   | PT1104504     | FIXED HODLER                   | 1   | 40  |
| 0.1   | PT1104505     | FIXED HODLER                   | 1   | 57  |
| 0.1   | PT1110612401  | VR KNOB PAINTING ABS+TPR       | 1   | 83  |
| 0.1   | PT11106129    | LAMPSHADE                      | 1   | 35  |
| 0.1   | PT1110613001  | ROTATE BUTTON PAINTING         | 1   | 19  |
| 0.1   | PT1110613701  | KNOB                           | 2   | 11  |
| 0.1   | PT11106151    | POWER BUTTON                   | 1   | 55  |
| 0.1   | PT11106178    | VR KNOB PAINITING ABS+ABS      | 1   | 13  |
| 0.1   | PT11106190    | LAMP(△ ▽)                      | 2   | 34  |
| 0.1   | PT1110619633  | KNOB (F/X)LASER                | 1   |     |

|     |                     |                                |          |                                                                                                     |
|-----|---------------------|--------------------------------|----------|-----------------------------------------------------------------------------------------------------|
| 0.1 | PT1110619634        | KNOB(TAP)LASER                 | 1        |                                                                                                     |
| 0.1 | PT111068006         | KNOB LASER                     | 3        | 18                                                                                                  |
| 0.1 | PT113020201         | CABINET PRINTING               | 1        | 32                                                                                                  |
| 0.1 | PW63200UC01         | AC POWER SUPPLY CORD           | 1        |                                                                                                     |
|     | UP301-104           | AC CORD(02)                    |          |                                                                                                     |
|     | PW53200UK03         | POWER SUPPLY CORD(03)          |          |                                                                                                     |
|     | PW43200SAA01        | POWER WIRE(04)                 |          |                                                                                                     |
|     | PW43200AR01         | POWER SUPPLY CORD(05)          |          |                                                                                                     |
|     | PW33200KS01         | POWER SUPPLY CORD(06)          |          |                                                                                                     |
|     | PW22200JIS02        | POWER WIRE(07)                 |          |                                                                                                     |
| 0.1 | RU111061501         | RUBBER KNOB(CUE/PLAY)LASER ETC | 1        | 25                                                                                                  |
| 0.1 | RU111061502         | RUBBER KNOB(PLAY)LASER ETCHING | 1        | 26                                                                                                  |
| 0.1 | RU111061503         | RUBBER KNOB(SYNC))LASER ETCHIN | 1        | 24                                                                                                  |
| 0.1 | RU111061601         | RUBBER KNOB(DELET)LASER ETCHIN | 1        | 23                                                                                                  |
| 0.1 | RU111061602         | RUBBER KNOB(ON/OFF)LASER ETCHI | 1        | 15                                                                                                  |
| 0.1 | RU111061612         | RUBBER KNOB LASER ECTING(MODE) | 1        | 14                                                                                                  |
| 0.1 | RU111061613         | RUBBER KNOB                    | 1        | 36                                                                                                  |
| 0.1 | RU111061701         | RUBBER KNOB(+/-)LASER ETCHING  | 1        | 33                                                                                                  |
| 0.1 | RU111061801         | RUBBER KNOB(ASSES KEY)LASER ET | 1        | 27                                                                                                  |
| 0.1 | RU111063001         | RUBBER KNOB (BACK/FWD)         | 1        | 70                                                                                                  |
| 0.1 | RU111063201         | RUBBER KNOB LASER ECTING       | 1        | 17                                                                                                  |
| 0.1 | RU111063202         | RUBBER KNOB LASER ECTING(1/2X) | 1        | 16                                                                                                  |
| 0.1 | RU1330701           | FOOT STAND                     | 4        | 71                                                                                                  |
| 0.1 | SC0305PIBI          | SCREW                          | 6        | 81                                                                                                  |
| 0.1 | SC0308ABBI          | SCREW(FLAT)                    | 30       | 10                                                                                                  |
| 0.1 | SC0308ABNI          | SCREW                          | 23       | 60                                                                                                  |
| 0.1 | SC0308AINI          | SCREW                          | 5        | 53                                                                                                  |
| 0.1 | SC0308PBBI          | SCREW M3x8mm PPB TAPPING       | 1        | 84                                                                                                  |
| 0.1 | SC0308RBNI          | SCREW M3*8MM PRN TAPPING       | 44       | 37                                                                                                  |
| 0.1 | SC0310WBNI          | SCREW                          | 6        | 6                                                                                                   |
| 0.1 | SC0333PIBI          | SCREW                          | 4        | 51                                                                                                  |
| 0.1 | SC0410RINI          | SCREW                          | 4        | 3                                                                                                   |
| 0.1 | SC2608W10BZI        | SCREW                          | 1        |                                                                                                     |
| 0.1 | SPSSNP-Y049-Y       | SWITCHING POWER(INTERNAL) 28V/ | 1        | 63                                                                                                  |
| 0.1 | TSNK14              | SOFTWARE CD                    | 1        |                                                                                                     |
| 0.1 | TWCD6030            | POWER SWITCH ASS'Y             | 1        | 58                                                                                                  |
| 0.1 | TWMT11044690101     | PLATTER PROECESS ASS'Y         | 1        | 4                                                                                                   |
| 0.1 | WS070409IN          | WASHER 4MM                     | 3        | 21                                                                                                  |
| 0.1 | WS073307IC          | WASHER FLAT                    | 2        | 89                                                                                                  |
| 0.1 | WS408604505B        | WASHER EXTERNAL                | 1        | 5                                                                                                   |
|     |                     |                                |          |                                                                                                     |
| 1   | <b>TWPC09C02201</b> | <b>MAIN PCB ASS'Y</b>          | <b>1</b> | <b>43</b>                                                                                           |
| 2   | AL0-17-0103         | RESISTOR ARRAY 4X10K 1/10W     | 2        | R25,26                                                                                              |
| 2   | AL0-17-0470         | RESISTOR ARRAY 4X47 1/10W      | 3        | R44,78,82                                                                                           |
| 2   | AL1-16-0472         | ELECT 47UF/35V (L)             | 4        | C11,73,87,100                                                                                       |
| 2   | AL2-50-4149         | DIODE 1N4148 100V 200mA        | 4        | D1,4,9,10                                                                                           |
| 2   | AL2-51-0340         | DIODE POWER SCHOTTKY MBRA340T3 | 1        | D6                                                                                                  |
| 2   | AL2-51-4003         | DIODE 300V 1A SMD              | 2        | D7,8                                                                                                |
| 2   | AL2-51-4401         | TRANSISTOR NPN 2N4401          | 1        | Q6                                                                                                  |
| 2   | AL2-51-4403         | TRANSISTOR PNP 2N4403          | 5        | Q1,7~10                                                                                             |
| 2   | AL2-61-0431         | REG LM431 2.5V-37V 100MA       | 1        | U15                                                                                                 |
| 2   | AL2-72-0339         | IC LM339                       | 1        | U13                                                                                                 |
| 2   | AL2-75-4345         | IC CS4345                      | 2        | U11,14                                                                                              |
| 2   | AL2-85-0403         | IC TPS60403                    | 1        | U18                                                                                                 |
| 2   | AL4-08-0402         | CON USB 4-PIN FEMALE PC-MNT HO | 1        | J4                                                                                                  |
| 2   | AL7-01-0046         | CRYSTAL 12MHz 49US SMD         | 1        | X1                                                                                                  |
| 2   | CS101J5003NPO       | CERAMIC 100P (SMD)             | 10       | C6~10,12~16                                                                                         |
| 2   | CS104K5003X7R       | CERAMIC 0.1UF (SMD)            | 49       | C1~5,17,19,21,22,24~29,36~38,40,47,48,50,56,57,60~62,67,70,71,74~78,81,83,84,86,90,91,95~98,103~106 |
| 2   | CS105K1603X5R       | CERAMIC 1UF (SMD)              | 4        | C18,107~109                                                                                         |

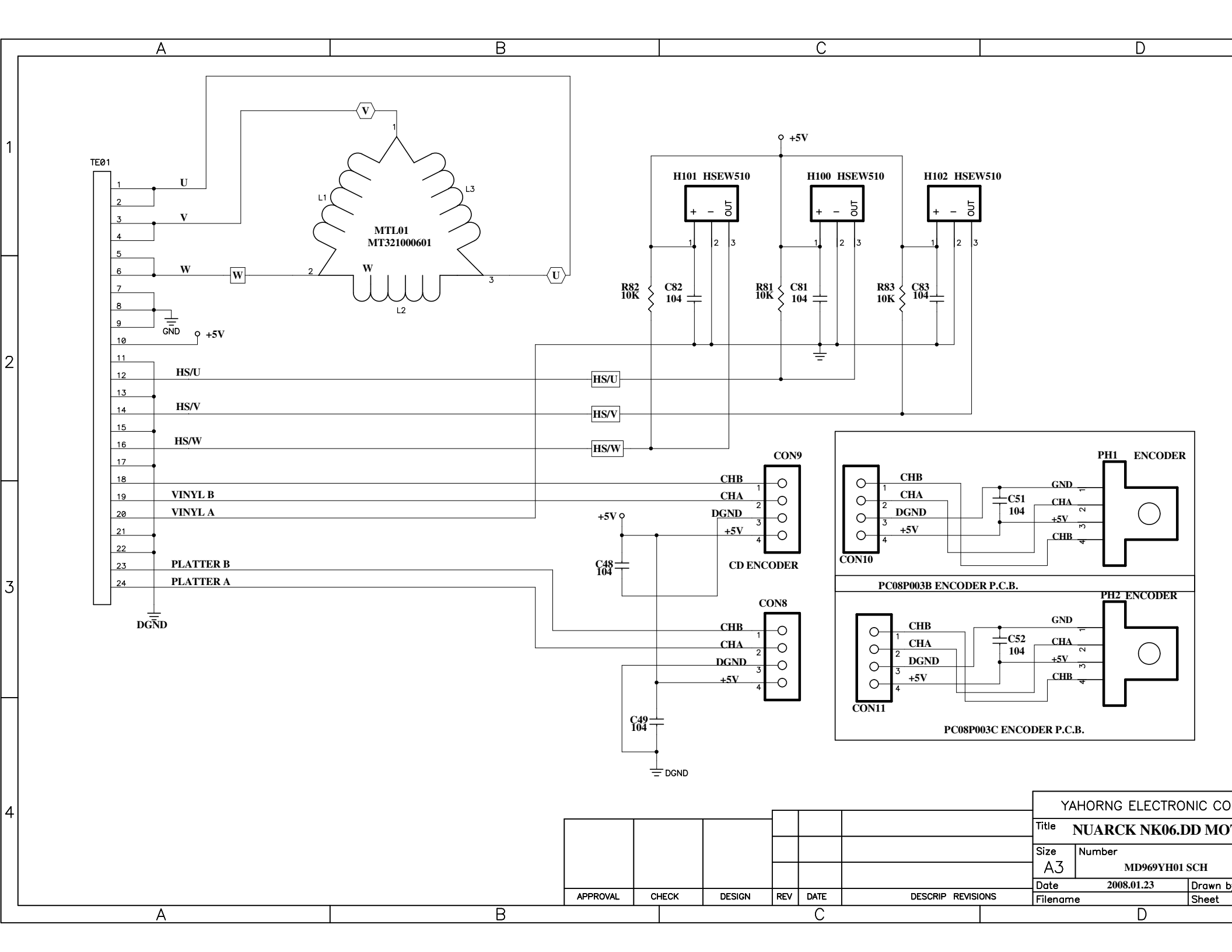
|   |                 |                                |    |                                            |
|---|-----------------|--------------------------------|----|--------------------------------------------|
| 2 | CS180J5003NPO   | CERAMIC 18P (SMD)              | 4  | C20,30,45,49                               |
| 2 | CS200J5003NPO   | CERAMIC 20P (SMD)              | 2  | C51,58                                     |
| 2 | DISDBAT54SSM    | DIODE BAT54S 30V 0.2A          | 2  | D2,5                                       |
| 2 | EC22650T        | ELECT 22UF/50V                 | 14 | C23,39,46,59,68,72,79,80,85,88,89,92,94,99 |
| 2 | EC22735TLE      | ELECT 220UF/35V LOW-ESR        | 4  | C69,82,93,101                              |
| 2 | EC47525TLE      | ELECT 4.7UF/25V LOW-ESR        | 1  | C35                                        |
| 2 | EC47625TNP      | ELECT 47UF/25V Nonpolar        | 8  | C31~34,63~66                               |
| 2 | EP350101        | TERMINAL (ψ3.5)                | 1  | J2                                         |
| 2 | FCN1T1003001    | 30P CONNECTOR                  | 1  | (J10)                                      |
| 2 | IC74AHCT541PWR  | IC 74AHCT541PWR                | 1  | U12                                        |
| 2 | ICAP1501K5L     | IC AP1501K5L                   | 1  | U10                                        |
| 2 | ICBA033FP       | IC BA033FP                     | 1  | U5                                         |
| 2 | ICISP1583BS     | IC ISP1583BS                   | 1  | U3                                         |
| 2 | ICLM317S        | IC LM317S                      | 3  | U9,16,17                                   |
| 2 | ICNJM-2114M     | IC NJM-2114M                   | 2  | U6,7                                       |
| 2 | ICSN75176AD     | IC SN75176AD                   | 2  | U1,2                                       |
| 2 | ICUSBLC62SC6    | IC USBLC62SC6                  | 1  | D3                                         |
| 2 | ICYHFR135       | IC YHFR135 (ATMEGA162)         | 1  | U4                                         |
| 3 | ICATMEGA162     | IC ATMEGA162                   | 1  | -                                          |
| 2 | ICYHFR136       | IC YHFR136 (LC4032V)           | 1  | U8                                         |
| 2 | ICLC4032V       | IC LC4032V                     | 1  | -                                          |
| 2 | JK08002-880SE03 | RJ-45 Jack 8Pin 90° Female     | 1  | J3                                         |
| 2 | MC33150T        | METALLIZED POLYESTER FILM CAPA | 4  | C52~55                                     |
| 2 | MC68150T        | MALYER CAPACITOR               | 4  | C41~44                                     |
| 2 | MT0701816       | FIX PLATE                      | 1  | (J1)                                       |
| 2 | MT1101810       | ground contact slice           | 1  | (J2)                                       |
| 2 | MT11044102      | FIXED PLATE                    | 1  | -                                          |
| 2 | PC09C022        | MAIN PCB                       | 1  | -                                          |
| 2 | RJ1107-09G      | JACK RCA                       | 1  | J1                                         |
| 2 | RS000004J06     | RESISTOR 0Ω (SMD)              | 2  | R47,89                                     |
| 2 | RS000010J03     | RESISTOR 0Ω (SMD)              | 5  | R9,77,85,90,92                             |
| 2 | RS001004J06     | RESISTOR 10Ω (SMD)             | 1  | R91                                        |
| 2 | RS001K10F03     | RESISTOR 1K (SMD)              | 4  | R13~16                                     |
| 2 | RS001K10F03     | RESISTOR 1K (SMD)              | 3  | R93,98,99                                  |
| 2 | RS001M10F03     | RESISTOR 1M (SMD)              | 8  | R1,2,5,6,11,65,105,106                     |
| 2 | RS004710F03     | RESISTOR 47Ω (SMD)             | 9  | R28,31,33,46,70~74                         |
| 2 | RS010004F06     | RESISTOR 100Ω (SMD)            | 2  | R7,32                                      |
| 2 | RS010010F03     | RESISTOR 100Ω (SMD)            | 1  | R94                                        |
| 2 | RS010K10F03     | RESISTOR 10K (SMD)             | 7  | R4,8,17,34~37                              |
| 2 | RS010K10F03     | RESISTOR 10K (SMD)             | 10 | R52~55,79~81,83,84,86                      |
| 2 | RS012K10F03     | RESISTOR 12K (SMD)             | 1  | R29                                        |
| 2 | RS01K110F03     | RESISTOR 1.1K (SMD)            | 1  | R87                                        |
| 2 | RS01K310F03     | RESISTOR 1.3K (SMD)            | 1  | R76                                        |
| 2 | RS01K510F03     | RESISTOR 1.5K (SMD)            | 1  | R30                                        |
| 2 | RS020K10F03     | RESISTOR 20K (SMD)             | 4  | R20~23                                     |
| 2 | RS024310F03     | RESISTOR 243Ω (SMD)            | 2  | R75,96                                     |
| 2 | RS03K410F03     | RESISTOR 3.4K (SMD)            | 4  | R61~64                                     |
| 2 | RS047010F03     | RESISTOR 470Ω (SMD)            | 1  | R60                                        |
| 2 | RS047K10F03     | RESISTOR 47K (SMD)             | 2  | R18,19                                     |
| 2 | RS04K710F03     | RESISTOR 4.7K (SMD)            | 8  | R40~43,66~69                               |
| 2 | RS060410F03     | RESISTOR 604Ω (SMD)            | 2  | R10,39                                     |
| 2 | RS082510F03     | RESISTOR 825Ω (SMD)            | 2  | R3,24                                      |
| 2 | RS100K10F03     | RESISTOR 100K (SMD)            | 7  | R12,27,100~104                             |
| 2 | RS16K910F03     | RESISTOR 16.9K (SMD)           | 5  | R56~59,88                                  |
| 2 | RS2K6710F03     | RESISTOR 2.67K (SMD)           | 1  | R95                                        |
| 2 | RS3K0910F03     | RESISTOR 3.09K (SMD)           | 1  | R97                                        |
| 2 | RS7K3210F03     | RESISTOR 7K32 (SMD)            | 4  | R48~51                                     |
| 2 | SC0306RBNI      | SCREW M3*6MM PRN TAPPING       | 1  | -                                          |
| 2 | SWVS6P001       | SWITCH                         | 2  | SW1,2                                      |
| 2 | TE512540801     | 8P CONNECTOR                   | 1  | (J7)                                       |

|   |                      |                                    |    |                                                                                                 |
|---|----------------------|------------------------------------|----|-------------------------------------------------------------------------------------------------|
| 2 | TE562540601          | 6P CONNECTOR                       | 1  | (J5)                                                                                            |
| 2 | TE612000202          | 2P CONNECTOR                       | 1  | (J6)                                                                                            |
| 2 | TE613960201          | 2P CONNECTOR                       | 1  | (J9)                                                                                            |
| 2 | TE613960401          | 4P CONNECTOR                       | 1  | (J8)                                                                                            |
| 2 | TR2SD2114KSM         | TRANSISTOR 2SD2114KSM (SMD) (2878) | 4  | Q2~5                                                                                            |
| 2 | UH160808T-102Y       | BEAD 1000Ω 100MHZ                  | 6  | L1~6                                                                                            |
| 2 | UHACM2012900-2P      | CHOKE 90Ω 0.4A 2.0X1.2mm           | 1  | L7                                                                                              |
| 2 | UHLFTBH1204-121      | COMMON MODE CHOKE,121UH,3A         | 1  | L10                                                                                             |
| 2 | UHSMTDRRI125-101M    | INDUCTANCE (100UH/1.3A)            | 1  | L8                                                                                              |
| 2 | UHSQD105T-751        | Common Inductor 700Ω ±35% 10       | 1  | L9                                                                                              |
| 2 | XTM11289X6           | CRYSTAL 11.2896MHz (SMD)           | 1  | X3                                                                                              |
| 2 | XTM16000X6           | CRYSTAL 16MHz (SMD)                | 1  | X2                                                                                              |
|   |                      |                                    |    |                                                                                                 |
| 1 | <b>TWPC09C02301</b>  | <b>FUNCTION PCB ASS'Y</b>          | 1  |                                                                                                 |
| 2 | <b>PC09C023</b>      | <b>FUNCTION PCB</b>                | 1  |                                                                                                 |
| 2 | <b>TWPC09C023A01</b> | <b>FUNCTION PCB ASS'Y</b>          | 1  | 66                                                                                              |
| 3 | AA821786X            | Oscillator 11.2896MHz              | 1  | X3                                                                                              |
| 3 | AL0-09-0044          | VARIABLE RESISTOR                  | 1  | P3                                                                                              |
| 3 | AL0-17-0103          | RESISTOR ARRAY 4X10K 1/10W         | 2  | R16,38                                                                                          |
| 3 | AL0-17-0470          | RESISTOR ARRAY 4X47 1/10W          | 11 | R40,64,71,73,83,86~89,109,118                                                                   |
| 3 | AL1-16-0472          | ELECT 47UF/35V (L)                 | 5  | C7,19,37,47,67                                                                                  |
| 3 | AL2-51-4401          | TRANSISTOR NPN 2N4401              | 1  | Q11                                                                                             |
| 3 | AL2-51-4403          | TRANSISTOR PNP 2N4403              | 2  | Q5,6                                                                                            |
| 3 | AL2-54-0138          | TRANSISTOR BSS138                  | 1  | Q2                                                                                              |
| 3 | AL2-72-0339          | IC LM339                           | 2  | U19,20                                                                                          |
| 3 | AL2-73-4053          | IC CD4053                          | 1  | U8                                                                                              |
| 3 | AL7-20-0075          | INDUCTANCE 2.2UH(SMD)              | 1  | L3                                                                                              |
| 3 | CS103K5003X7R        | CERAMIC 0.01UF (SMD)               | 4  | C14,15,44,50                                                                                    |
| 3 | CS104K5003X7R        | CERAMIC 0.1UF (SMD)                | 43 | C1~4,11~13,17,20~28,30~34,38~42,46,48,49,52,54,55,90,91,94,96,103,137,138,142,144,146           |
| 3 | CS182J5005NPO        | CERAMIC 1.8nF (SMD)                | 1  | C16                                                                                             |
| 3 | CS220J5003NPO        | CERAMIC 22P (SMD)                  | 2  | C18,43                                                                                          |
| 3 | EC47525SM            | ELECT 4.7UF/25V (SMD)              | 3  | C29,35,51                                                                                       |
| 3 | FCN1S1000401         | 4P FFC CONNECTOR                   | 1  | (J7)                                                                                            |
| 3 | FCN1S1251602         | 16P FFC CONNECTOR                  | 1  | (J2)                                                                                            |
| 3 | FCN1T1003001         | 30P FFC CONNECTOR                  | 1  | (J3)                                                                                            |
| 3 | IC74ACT574PW         | IC 74ACT574PW                      | 9  | U1~3,6,7,9,10,21,24                                                                             |
| 3 | IC74AHC245PW         | IC SN74AHC245PWR                   | 1  | U30                                                                                             |
| 3 | ICBA033FP            | IC BA033FP                         | 1  | U38                                                                                             |
| 3 | ICLM317S             | IC LM317S                          | 1  | U16                                                                                             |
| 3 | ICYHFRC137           | IC YHFRC137 (AL2-70-3433)          | 1  | U5                                                                                              |
| 4 | AL2-70-3433          | AL2-70-3433                        | 1  | -                                                                                               |
| 3 | ICYHFRC138           | IC YHFRC138 (AL2-80-0256)          | 1  | U4                                                                                              |
| 4 | AL2-80-0256          | AL2-80-0256                        | 1  | -                                                                                               |
| 3 | LD-005HLV            | RED LED                            | 50 | LED1~9,12,13,15,17~20,22,23,28~31,33,37,38~42,44,45,47,49,51,52,56,61,64,65,68~73,78,80,105~107 |
| 3 | LD304W2CB            | WHITE LED                          | 22 | LED10,11,14,16,21,24,26,27,34~36,43,46,48,50,53~55,57,66,67,82                                  |
| 3 | LDKPT-2012SECJ3      | RED LED (SMD)                      | 11 | LED79,81,83~91                                                                                  |
| 3 | LDYH-11R5W1R5        | LED BAR                            | 1  | (LED25,32,58~60,62,63,74~77)                                                                    |
| 3 | MT070446701          | IRON PLATE                         | 2  | (E1,3)                                                                                          |
| 3 | MT1104473            | FIXED PLATE                        | 2  | (P12)                                                                                           |
| 3 | NM3506103            | VELET CARPET                       | 1  | (P5)                                                                                            |
| 3 | PT111048005          | FIXED BASE,LED                     | 72 | (LED1~24,26~31,33~57,61,64~73,78,80,82,105~107)                                                 |
| 3 | RS000008J05          | RESISTOR 0Ω (SMD)                  | 1  | R150                                                                                            |
| 3 | RS000010J03          | RESISTOR 0Ω (SMD)                  | 3  | C86,R124,135                                                                                    |
| 3 | RS001K10F03          | RESISTOR 1K (SMD)                  | 4  | R39,77,126,145                                                                                  |
| 3 | RS004710F03          | RESISTOR 47Ω (SMD)                 | 5  | R84,85,90,167,168                                                                               |
| 3 | RS010K10F03          | RESISTOR 10K (SMD)                 | 11 | R3,4,9,13,32,80,81,143,157,161,162                                                              |
| 3 | RS018010F03          | RESISTOR 180Ω (SMD)                | 15 | R41,54,65,66,169~179                                                                            |
| 3 | RS01K810F03          | RESISTOR 1.8K (SMD)                | 1  | R12                                                                                             |

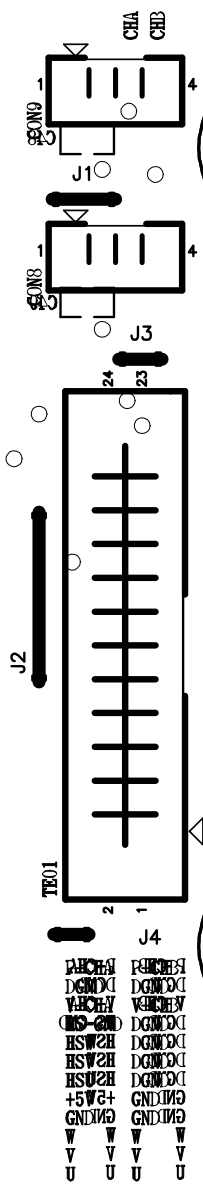
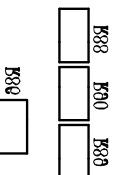
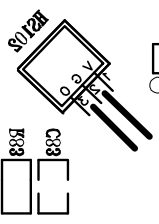
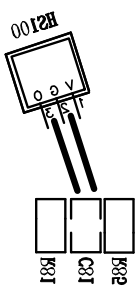
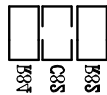
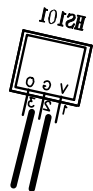
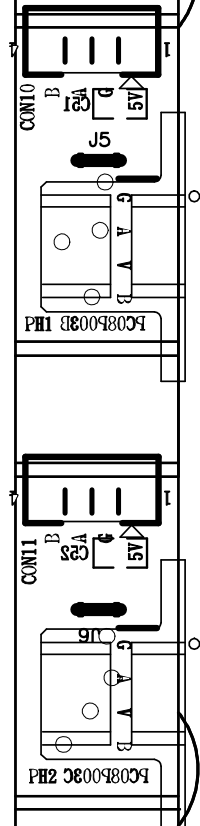


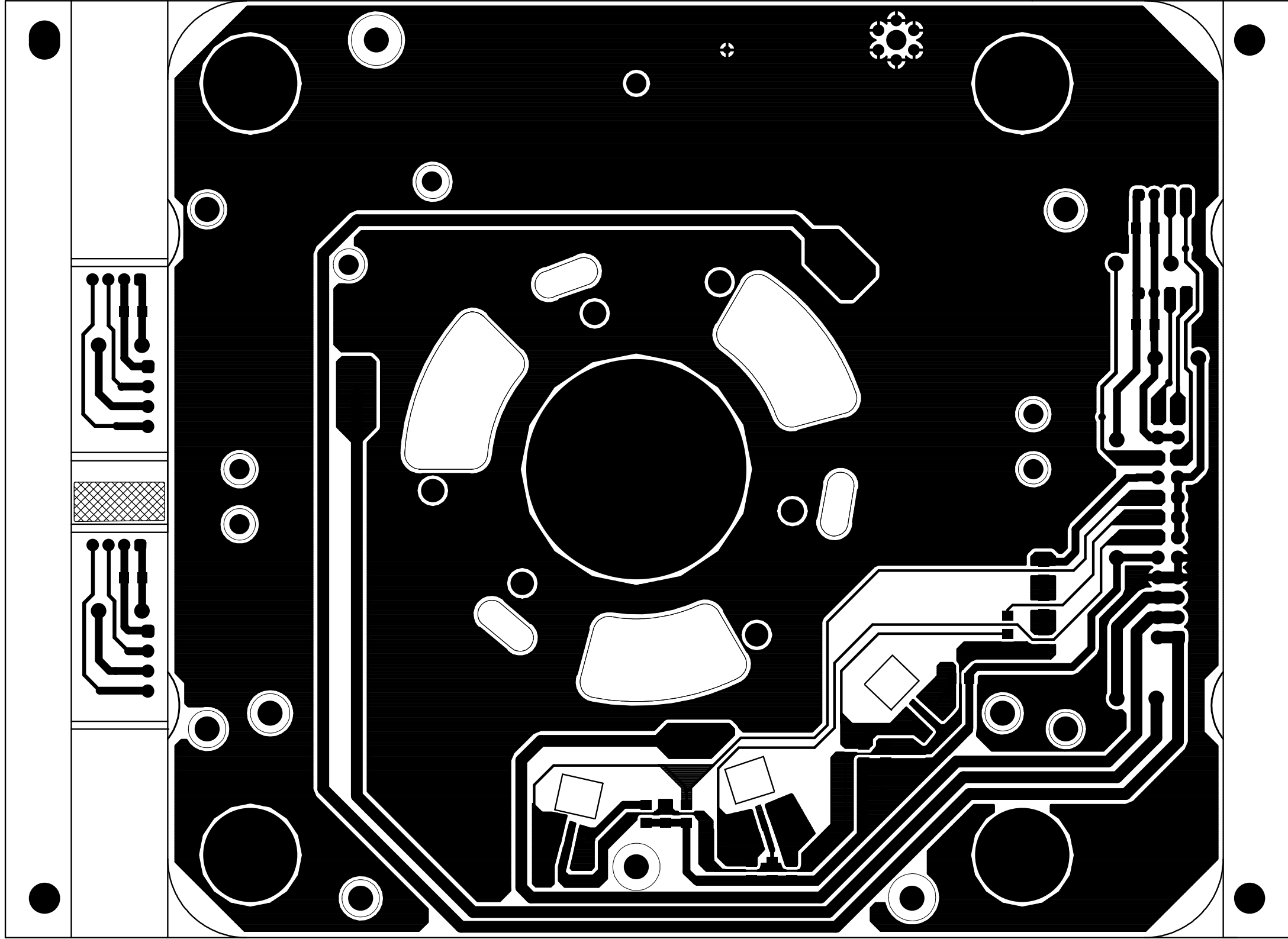
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| 3 | RS020010F03          | RESISTOR 200Ω (SMD)            | 1        | R146                                                                        |
| 3 | RS020K10F03          | RESISTOR 20K (SMD)             | 6        | R159,160,180~183                                                            |
| 3 | RS022010F03          | RESISTOR 220Ω (SMD)            | 28       | R14,17,18,51,69,72,74,75,78,82,92~95,98,100~105,110,117,133,136,138,140,142 |
| 3 | RS027010F03          | RESISTOR 270Ω (SMD)            | 12       | R31,33,37,49,55~57,59~61,128,131                                            |
| 3 | RS033010F03          | RESISTOR 330Ω (SMD)            | 16       | R50,67,68,70,76,96,97,99,107,112,129,132,134,137,139,141                    |
| 3 | RS03K310F03          | RESISTOR 3.3K (SMD)            | 11       | R79,91,106,108,113~115,119,121,123,127                                      |
| 3 | RS047010F03          | RESISTOR 470Ω (SMD)            | 25       | R8,11,19,25~30,42~46,48,111,116,120,125,151~156                             |
| 3 | RS056010F03          | RESISTOR 560Ω (SMD )           | 2        | R122,149                                                                    |
| 3 | RS100K10F03          | RESISTOR 100K (SMD)            | 6        | R23,47,52,53,58,147                                                         |
| 3 | RS1K6210F03          | RESISTOR 1.62K (SMD)           | 1        | R144                                                                        |
| 3 | RS22K110F03          | RESISTOR 22.1K (SMD)           | 1        | R130                                                                        |
| 3 | SC2606RINI           | SCREW                          | 2        | (P12)                                                                       |
| 3 | SWDC11B2021431       | SWITCH                         | 2        | E1,3                                                                        |
| 3 | SWEVQ11L05R          | Switch Tact Stem Height 5mm Op | 35       | S1~8,10~12,14~17,20~31,42~48,65                                             |
| 3 | SWSSFUB22-10K        | Switch Slide Single Row 180°   | 1        | S9                                                                          |
| 3 | TE512540801          | 8P CONNECTOR                   | 1        | (J1)                                                                        |
| 3 | TE962540801          | 8P CONNECTOR                   | 1        | (J505)                                                                      |
| 3 | TE962541401          | 14P CONNECTOR                  | 1        | (J16)                                                                       |
| 3 | VRR10301301          | Potentiometer Rotary 10k Shaft | 2        | P1,2                                                                        |
| 3 | VRS10303006          | Potentiometer Slide 10k 15mm   | 1        | P12                                                                         |
| 3 | VRS10310013          | Potentiometer Rotary Valume 10 | 1        | P5                                                                          |
| 3 | XTM40000X4           | Crystal 40MHz 49US DIP Fundame | 1        | X1                                                                          |
|   |                      |                                |          |                                                                             |
| 2 | <b>TWPC09C023B01</b> | <b>Bleep/Reverse PCB ASSY</b>  | <b>1</b> | <b>28</b>                                                                   |
| 3 | SWLBCR23M-095        | SWITCH                         | 1        | S49                                                                         |
| 3 | TE562540803          | 8P CONNECTOR                   | 1        | (J1)                                                                        |
|   |                      |                                |          |                                                                             |
| 2 | <b>TWPC09C023C01</b> | <b>SHIELD PCB ASS'Y</b>        | <b>1</b> |                                                                             |
|   |                      |                                |          |                                                                             |
| 1 | <b>TWPC09P00801</b>  | <b>MOTOR CONTRON PCB ASS'Y</b> | <b>1</b> | <b>61</b>                                                                   |
| 2 | CC104100Z            | CERAMIC 0.1UF/100V             | 1        | C39                                                                         |
| 2 | CS102K5005X7R        | CERAMIC 1nF (SMD)              | 3        | C9,41,42                                                                    |
| 2 | CS103K5005X7R        | CERAMIC 0.01UF (SMD)           | 1        | C54                                                                         |
| 2 | CS104Z5005Y5V        | CERAMIC 0.1UF (SMD)            | 26       | C1,2,7,8,14,16~26,32~35,37,44,48,57,58,59                                   |
| 2 | CS200J5005NPO        | CERAMIC 20P (SMD)              | 2        | C5,6                                                                        |
| 2 | CS334K1605X7R        | CERAMIC 0.33UF (SMD)           | 1        | C3                                                                          |
| 2 | CS682K5005X7R        | CERAMIC 6.8nF (SMD)            | 1        | C4                                                                          |
| 2 | DIN5406              | DIODE DIN5406                  | 1        | D3                                                                          |
| 2 | DISR26               | SCHOTTKY BARRIER RECTIFIER     | 1        | D01                                                                         |
| 2 | DIZDP6SMB47CAT3      | DIODE TVS (47V)                | 2        | ZD01,02                                                                     |
| 2 | EC10616T             | ELECT 10UF/16V                 | 2        | C12,15                                                                      |
| 2 | EC10716T             | ELECT 100UF/16V                | 1        | C36                                                                         |
| 2 | EC10850              | ELECT 1000UF/50V               | 1        | C38                                                                         |
| 2 | EC33725TLE           | ELECT 330UF/25V LOW-ESR        | 1        | C30                                                                         |
| 2 | EC33750T             | ELECT 330UF/50V                | 1        | C11                                                                         |
| 2 | EC47550T             | ELECT 4.7UF/50V                | 3        | C27,28,29                                                                   |
| 2 | EC47616T             | ELECT 47UF/16V                 | 1        | C13                                                                         |
| 2 | FCN1S1251602         | 16P CONNECTOR                  | 1        | CON02                                                                       |
| 2 | IC74AHC245PW         | IC 74AHC245PWR                 | 1        | U2                                                                          |
| 2 | ICAP1501K5L          | IC AP1501K5L                   | 1        | IC2                                                                         |
| 2 | ICAPL1117-33V        | IC APL1117-33V                 | 1        | IC3                                                                         |
| 2 | ICBA4558F            | IC BA4558F                     | 1        | IC8                                                                         |
| 2 | ICMIC809             | IC MIC809                      | 1        | IC4                                                                         |
| 2 | ICOPA354             | IC OPA354                      | 1        | IC1                                                                         |
| 2 | ICYH08P01            | IC YH08P01 (ICTMS320LF2407A)   | 1        | IC5                                                                         |
| 3 | ICTMS320LF2407A      | IC TMS320LF2407A               | 1        | -                                                                           |
| 2 | MOF01003             | M.RES 1Ω 3W                    | 1        | R47                                                                         |
| 2 | MT0703321            | FIXED PLATE                    | 2        | -                                                                           |

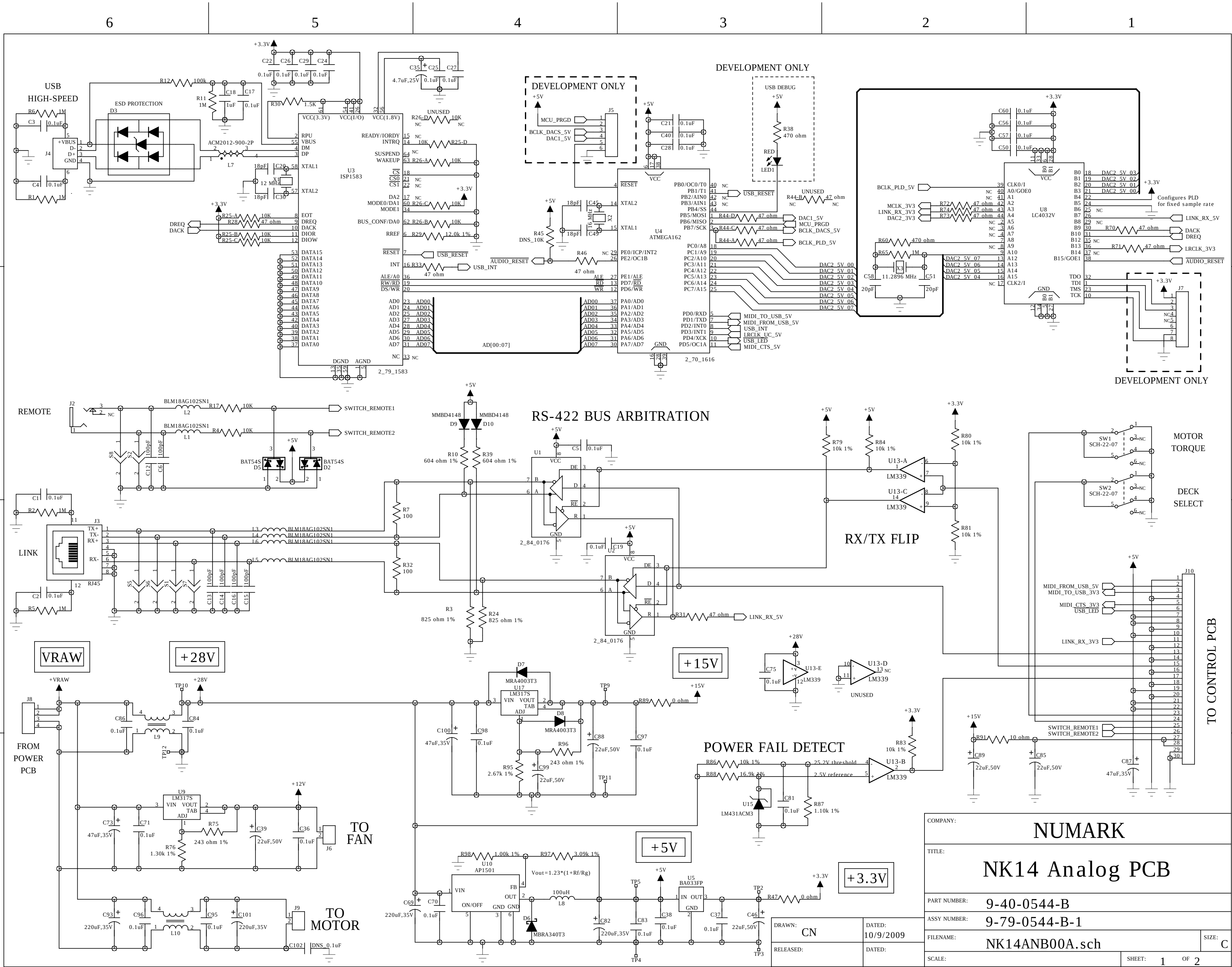
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|---|----------------|-----------------------------------|----|--------------------------------------------------------------------|
| 2 | NP211MAR79     | NAME PLATE                        | 1  | -                                                                  |
| 2 | PC09P008       | MOTOR CONTRON                     | 1  | -                                                                  |
| 2 | R10112         | RESISTOR 100Ω                     | 3  | R20,29,38                                                          |
| 2 | RS000004J06    | RESISTOR 0Ω (SMD)                 | 1  | R11                                                                |
| 2 | RS001608F05    | RESISTOR 16Ω (SMD)                | 1  | R8                                                                 |
| 2 | RS001K08F05    | RESISTOR 1K (SMD)                 | 18 | R14,17,19,24,28,32,37,46,51,55,59,63,68~70,75,115,118              |
| 2 | RS001M08F05    | RESISTOR 1M (SMD)                 | 1  | R21                                                                |
| 2 | RS010K08F05    | RESISTOR 10K (SMD)                | 30 | R1~7,9,10,12,16,22,26,30,31,35,40,44,48,53,56,61,94~98,110,112,116 |
| 2 | RS02K208F05    | RESISTOR 2.2K (SMD)               | 2  | R93,114                                                            |
| 2 | RS033K08F05    | RESISTOR 33K (SMD)                | 4  | R78,79,86,91                                                       |
| 2 | RS03K308F05    | RESISTOR 3.3K (SMD)               | 1  | R117                                                               |
| 2 | RS03K908F05    | RESISTOR 3K9 (SMD)                | 6  | R13,23,33,42,50,58                                                 |
| 2 | RS04K708F05    | RESISTOR 4.7K (SMD)               | 6  | R15,25,34,43,52,60                                                 |
| 2 | RS06K808F05    | RESISTOR 6.8K (SMD)               | 7  | R18,27,36,45,54,62,111                                             |
| 2 | RS100K08F05    | RESISTOR 100K 1% (SMD)            | 1  | R113                                                               |
| 2 | RS1K0508F05    | RESISTOR 1.05K (SMD)              | 1  | R65                                                                |
| 2 | RS3K2408F05    | RESISTOR 3.24K (SMD)              | 1  | R64                                                                |
| 2 | SC0308RBNI     | SCREW M3*8MM PRN TAPPING          | 6  | -                                                                  |
| 2 | SC3508RBNI     | SCREW                             | 4  | -                                                                  |
| 2 | TE613960201    | 2P CONNECTOR                      | 1  | CON7                                                               |
| 2 | TE962541401    | 14P CONNECTOR                     | 1  | CON1                                                               |
| 2 | TE962542401    | 24P CONNECTOR                     | 1  | CON3                                                               |
| 2 | TR2SA1037ASM   | TRANSISTOR 2SA1037ASM (SMD) (608) | 1  | Q43                                                                |
| 2 | TR2SB1340-T-B  | TRANSISTOR 2SB1340-T-B (1340)     | 3  | Q6,11,16                                                           |
| 2 | TR2SC2412KSM   | TRANSISTOR 2SC2412KSM (SMD) (536) | 1  | Q39                                                                |
| 2 | TR2SD1889      | TRANSISTOR 2SD1889                | 3  | Q23,28,33                                                          |
| 2 | TRIIRML5103    | TRANSISTOR IRLML5103              | 6  | Q4,9,14,21,26,31                                                   |
| 2 | TRRK7002       | TRANSISTOR RK7002                 | 18 | Q2,3,5,7,8,10,12,13,15,19,20,22,24,25,27,29,30,32                  |
| 2 | UH322522T-100K | CHOKE COIL 100UH                  | 2  | L1,2                                                               |
| 2 | UH453232T-3R3K | CAPACITOR                         | 2  | L4,5                                                               |
| 2 | UHPID1207-680M | Inductor 68uH±20% 3A SMD          | 1  | L3                                                                 |
| 2 | XTM08000X6F1   | CRYSTAL 8MHz 49US (SMD)           | 1  | Y2                                                                 |
|   |                |                                   |    |                                                                    |
| 1 | MD-969YH01     | DD MOTOR                          | 2  |                                                                    |
| 2 | TWPC08P00301   | MOTOR PCB ASS'Y                   | 1  |                                                                    |
| 2 | PC08P003       | MOTOR PCB SINGLE LAYER FR1(146    | 1  |                                                                    |
| 3 | TWPC08P003A01  | MOTOR PCB ASS'Y                   | 1  |                                                                    |
| 4 | CS104Z5005Y5V  | CEAMIC 0.1UF (SMD)                | 5  | C48,49,81,82,83                                                    |
| 4 | HSEW510        | HALL SENSOR IC EW510              | 3  | HS100,101,102                                                      |
| 4 | JW5206T        | JUMP WIRE                         | 4  | J1,2,3,4                                                           |
| 4 | RS000008J05    | RESISTOR 0Ω (SMD)                 | 7  | R84,85,86,87,88,89,90                                              |
| 4 | RS010K08F05    | RESISTOR 10K (SMD)                | 3  | R81,82,83                                                          |
| 4 | TE612000402    | 4P CONNECTOR                      | 1  | CON8                                                               |
| 4 | TE612000403    | 4P CONNECTOR                      | 1  | CON9                                                               |
| 4 | TE962542401    | 24P CONNECTOR                     | 1  | TE01                                                               |
|   |                |                                   |    |                                                                    |
| 3 | TWPC08P003B01  | OPTICAL ENCODER PCB ASS'Y         | 1  |                                                                    |
| 4 | CN0420000704   | 4P CONNECTOR (70mm)               | 1  | CON10                                                              |
| 4 | CS104Z5005Y5V  | CERAMIC 0.1UF (SMD)               | 1  | C51                                                                |
| 4 | DIHEDS-9700F50 | SENSOR                            | 1  | PH1                                                                |
| 4 | JW5206T        | JUMP WIRE                         | 1  | J5                                                                 |
|   |                |                                   |    |                                                                    |
| 3 | TWPC08P003C01  | OPTICAL ENCODER PCB ASS'Y         | 1  |                                                                    |
| 4 | CN0420001302   | 4P CONNECTOR (130mm)              | 1  | CON11                                                              |
| 4 | CS104Z5005Y5V  | CERAMIC 0.1UF (SMD)               | 1  | C52                                                                |
| 4 | DIHEDS-9700F50 | SENSOR                            | 1  | PH2                                                                |
| 4 | JW5206T        | JUMP WIRE                         | 1  | J6                                                                 |



PC08P003A PC08P003B  
2008.06.03 2008.08.00S



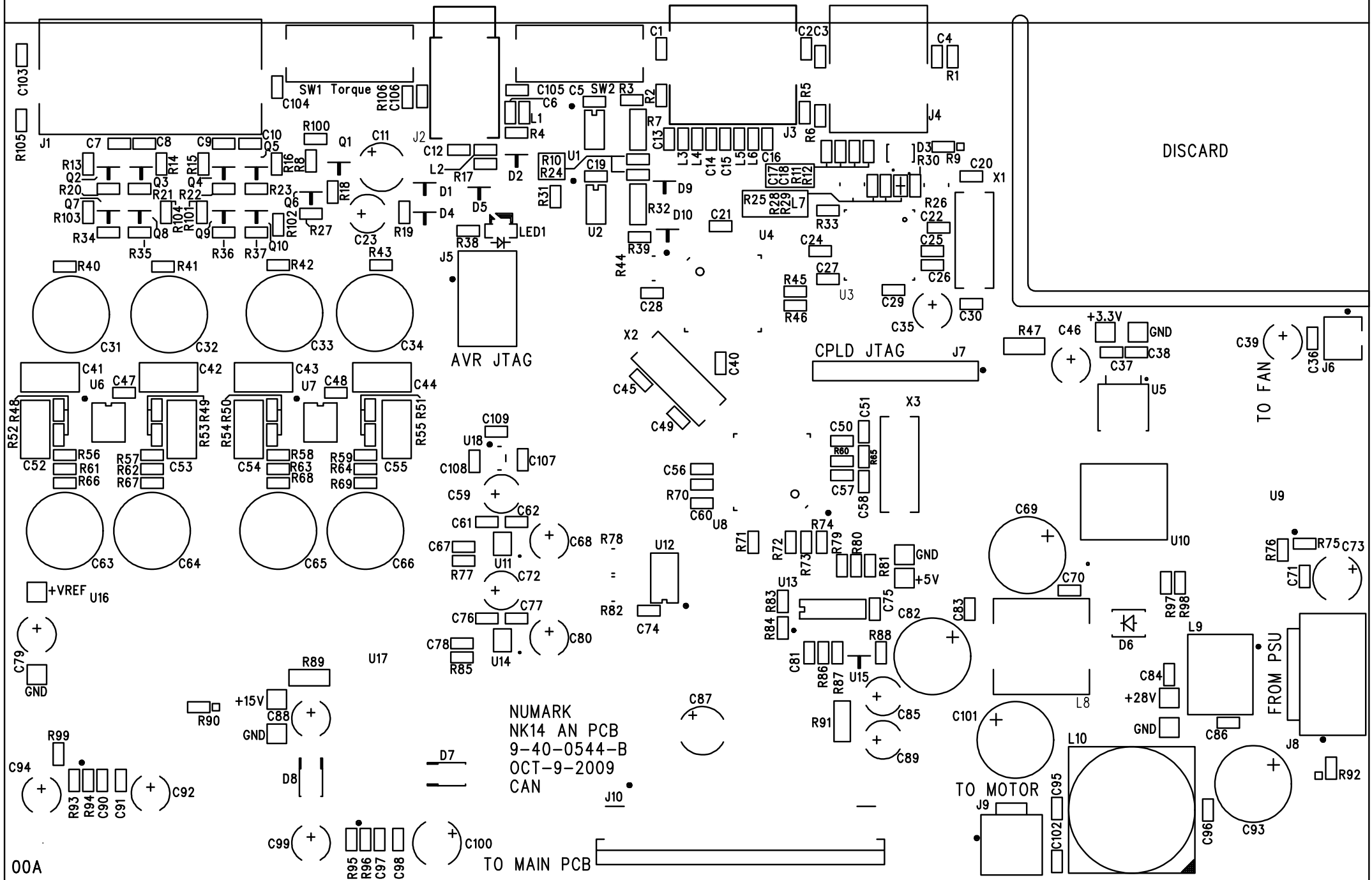




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|--------------|--|------------------|--|
| COMPANY:     |  | NUMARK           |  |
| TITLE:       |  | NK14 Analog PCB  |  |
| PART NUMBER: |  | 9-40-0544-B      |  |
| ASSY NUMBER: |  | 9-79-0544-B-1    |  |
| FILENAME:    |  | NK14ANB00A.sch   |  |
| SCALE:       |  | SHEET: 1 OF 2    |  |
| DRAWN: CN    |  | DATED: 10/9/2009 |  |
| RELEASED:    |  | DATED:           |  |

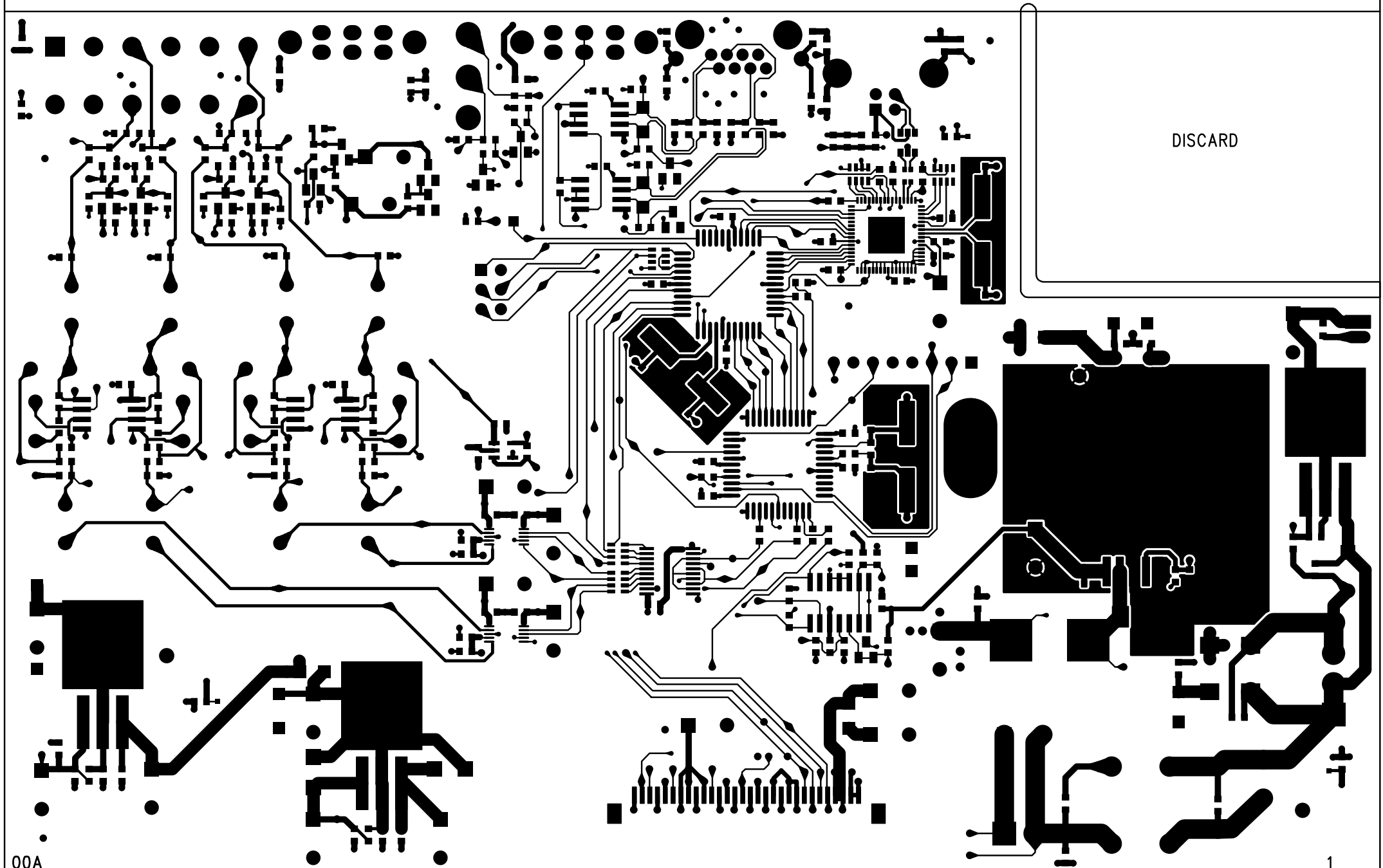


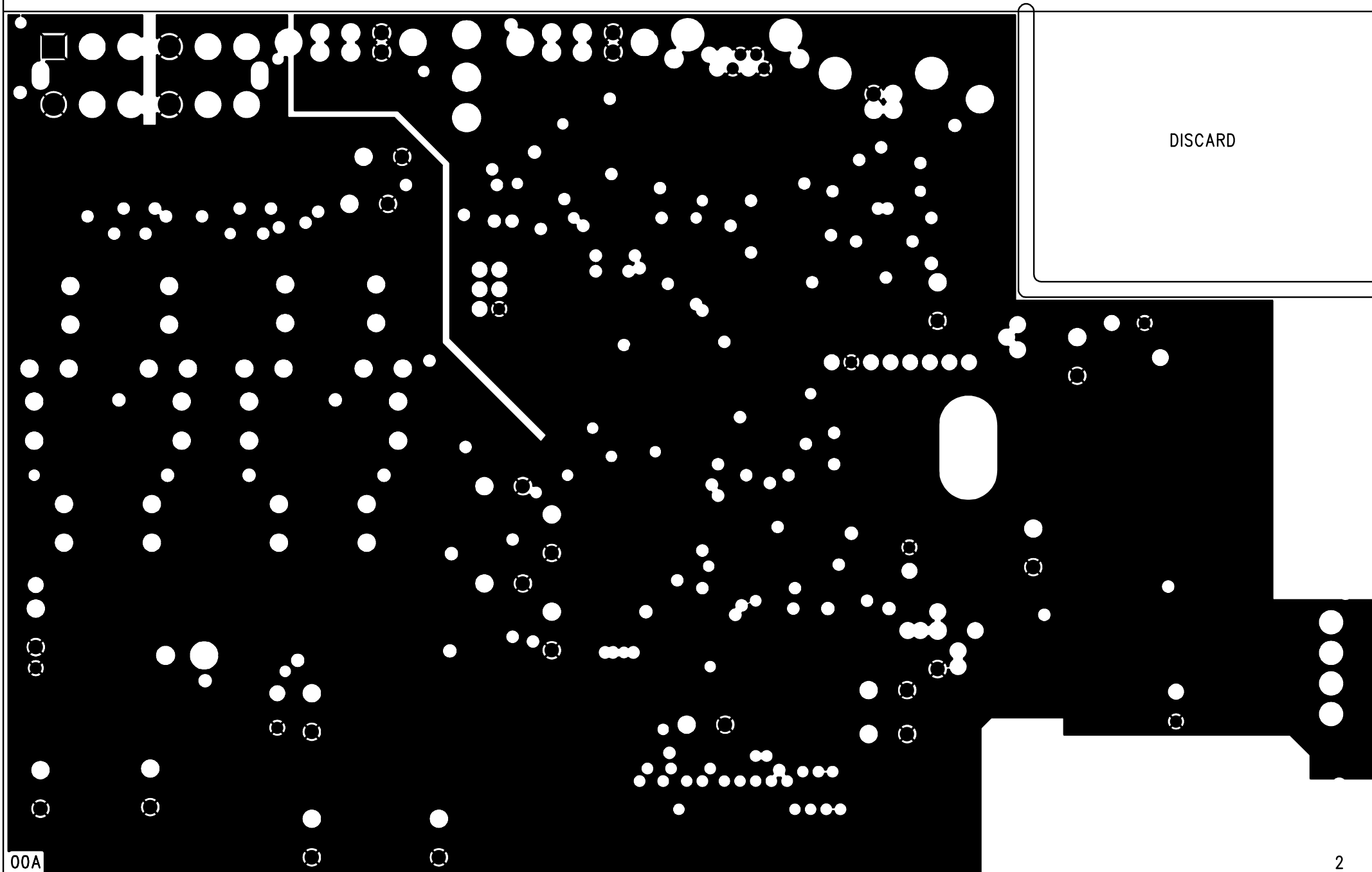




NUMARK  
NK14 AN PCB  
9-40-0544-B  
OCT-9-2009  
CAN



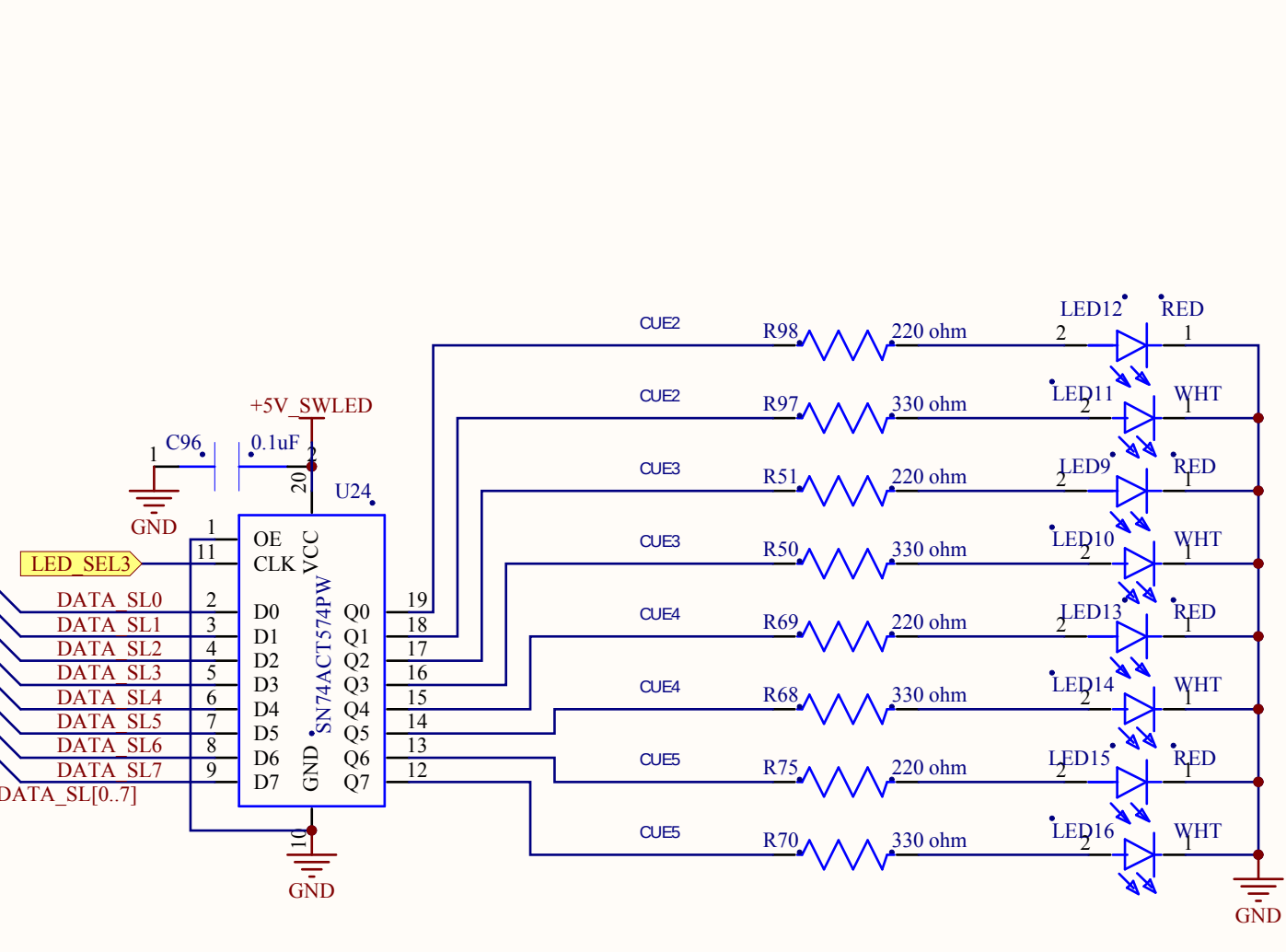
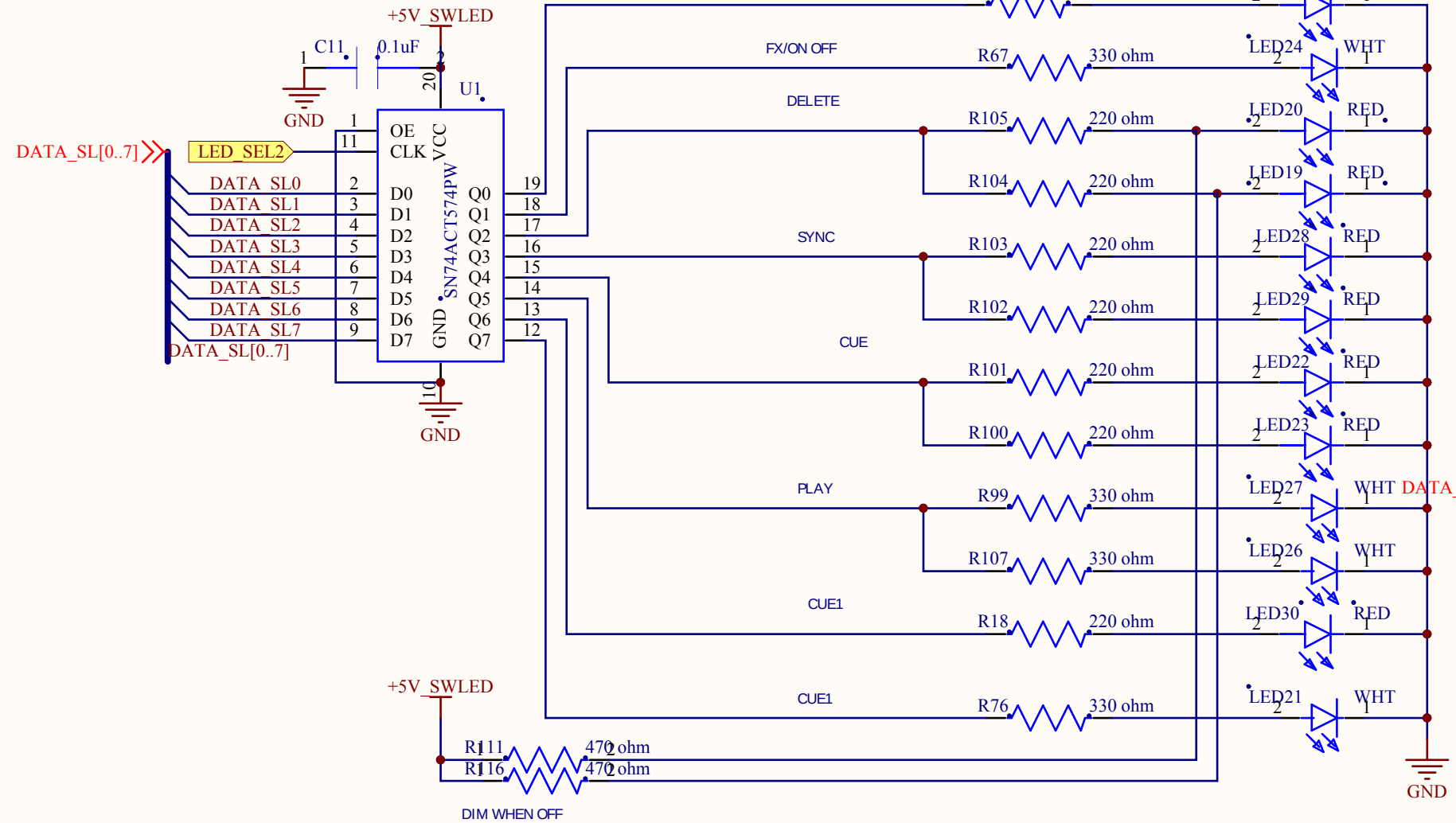
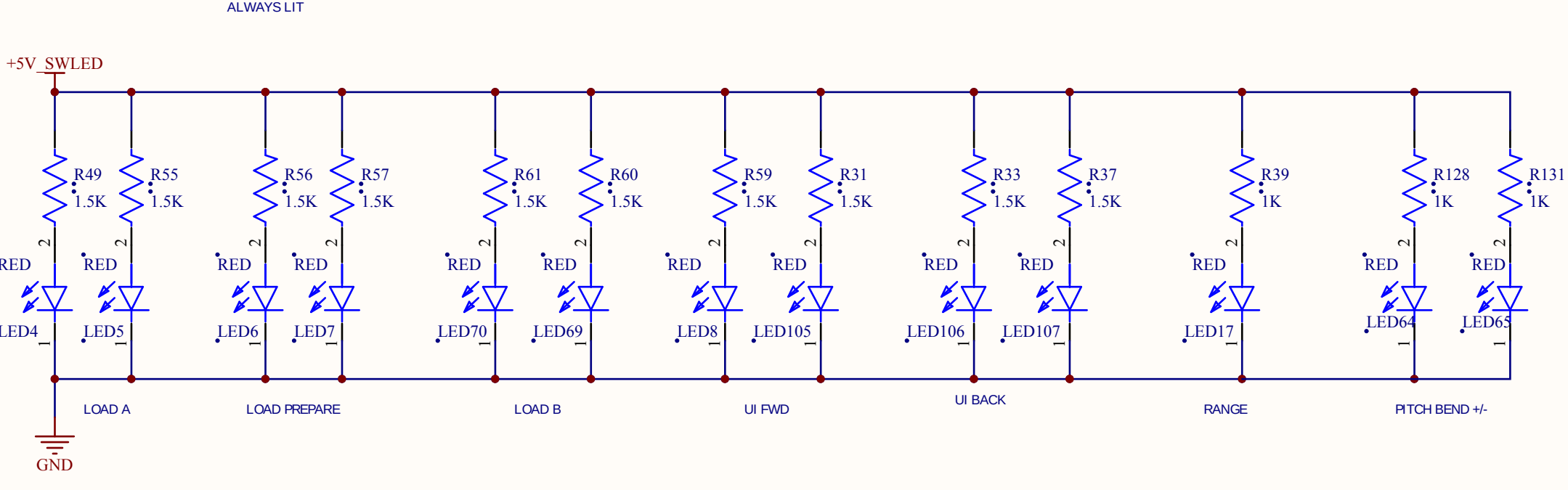
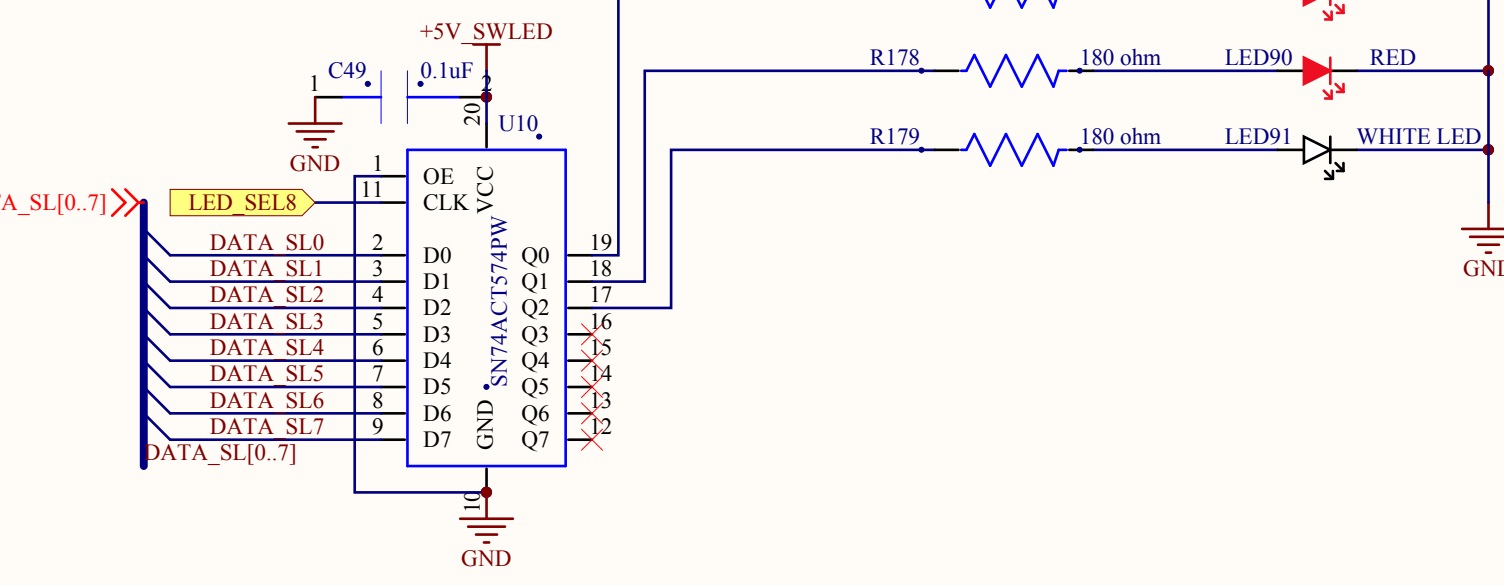
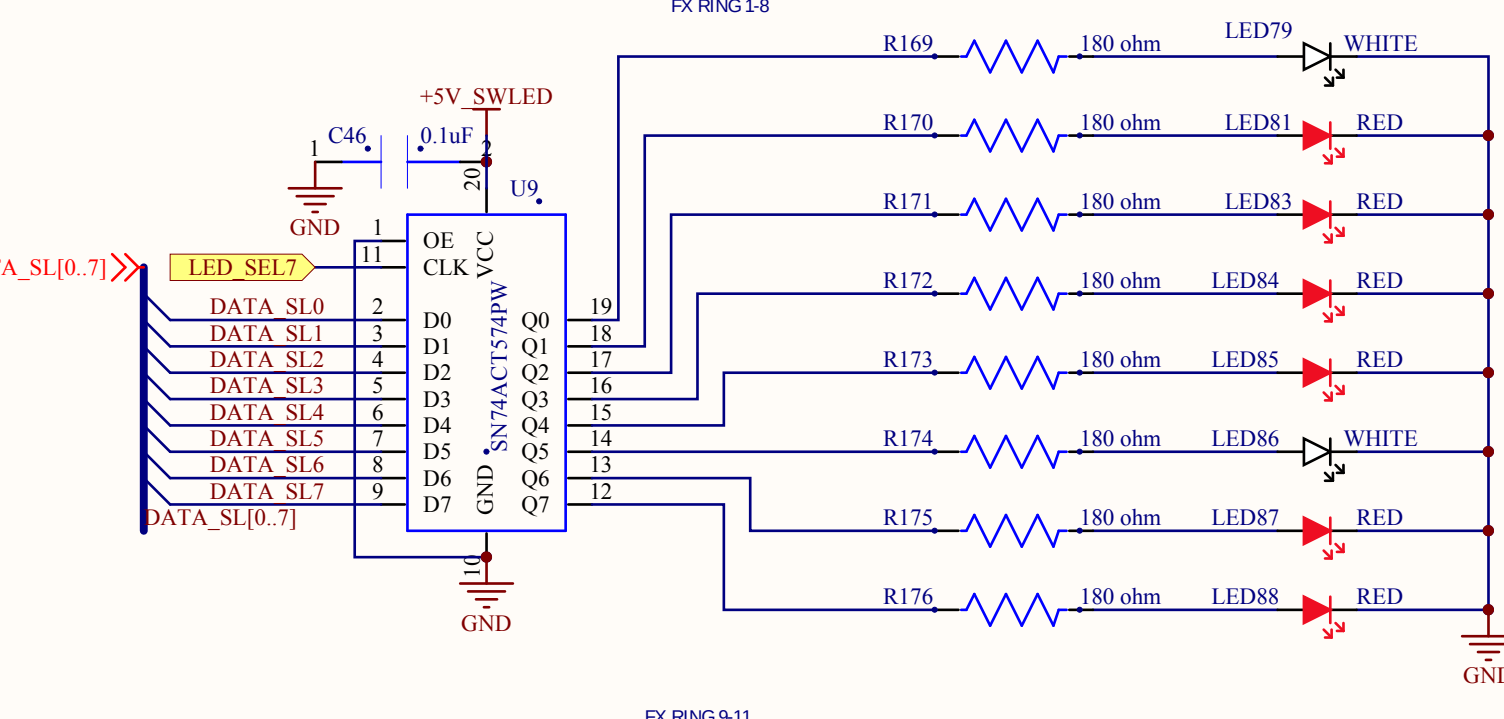
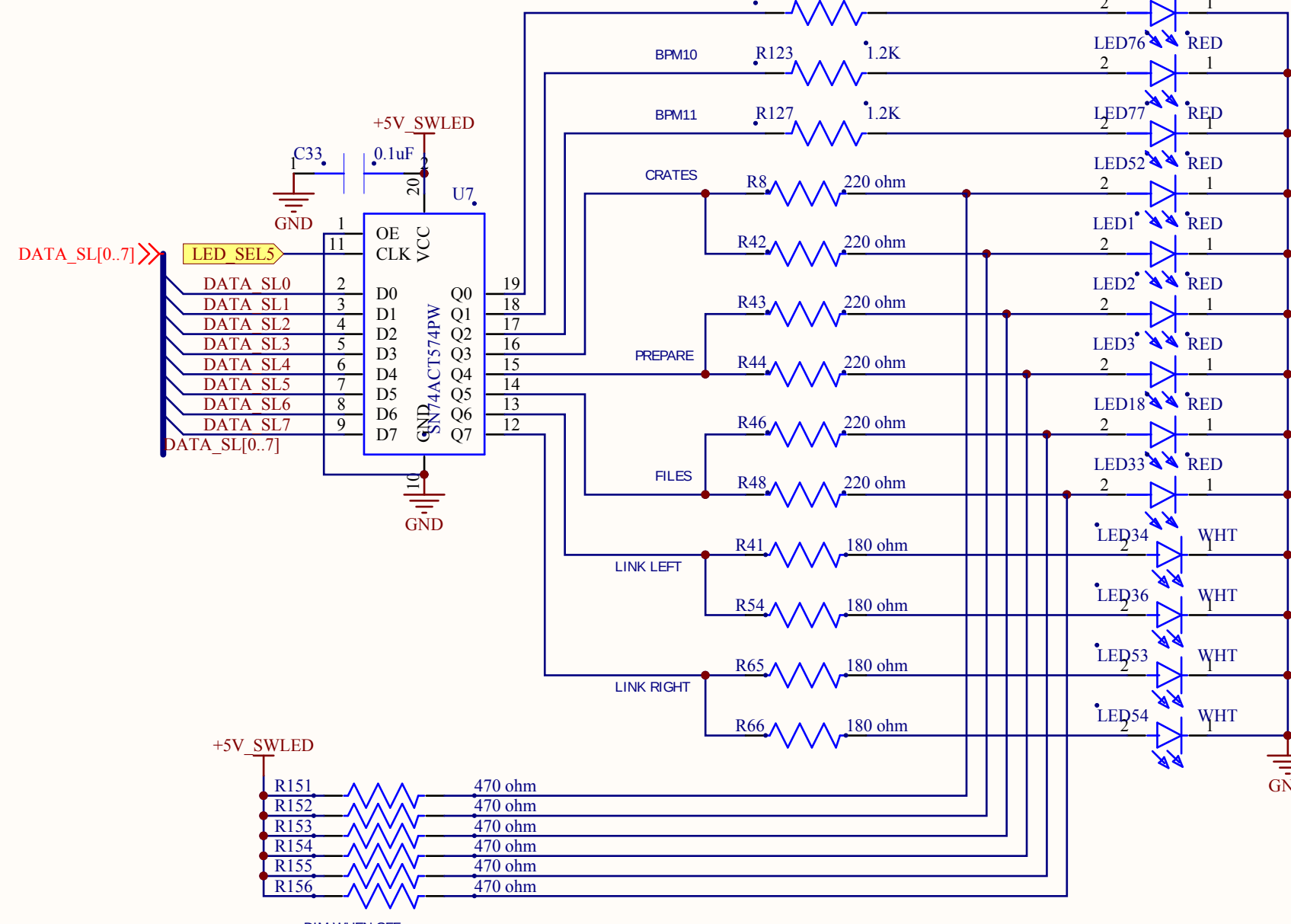
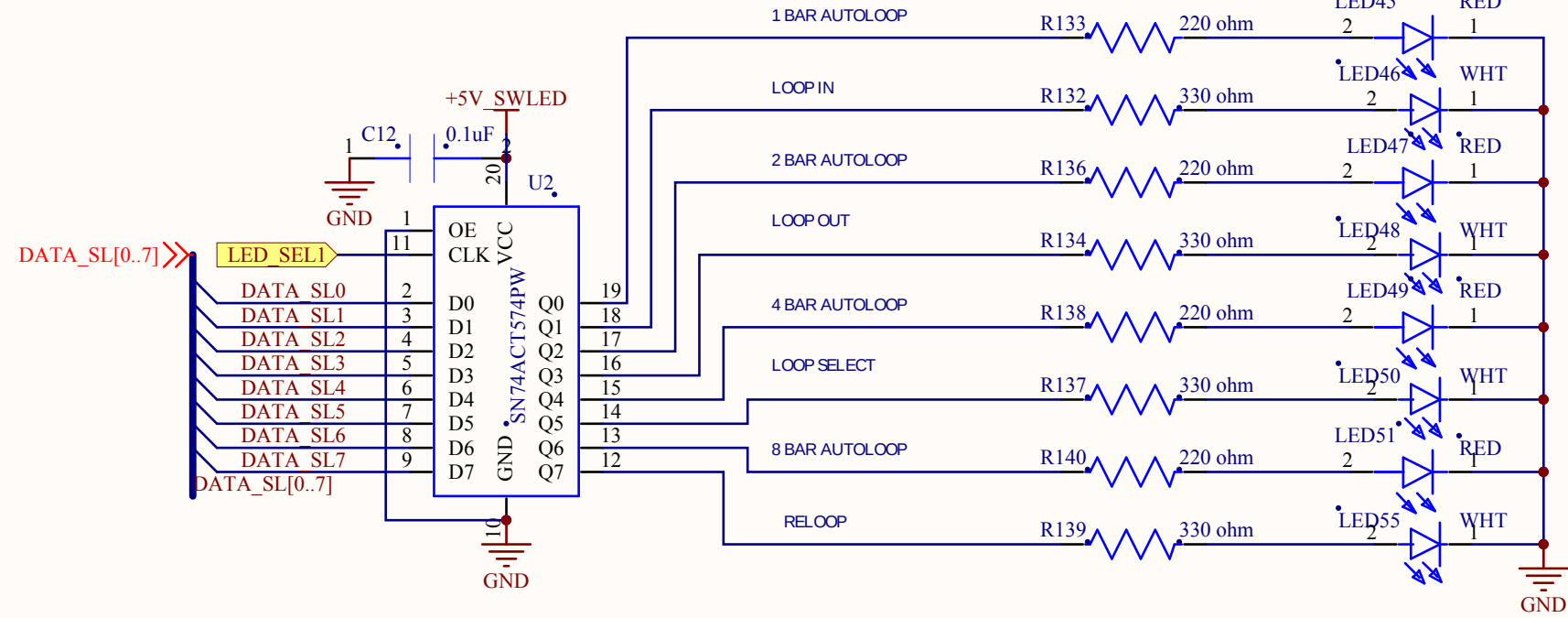
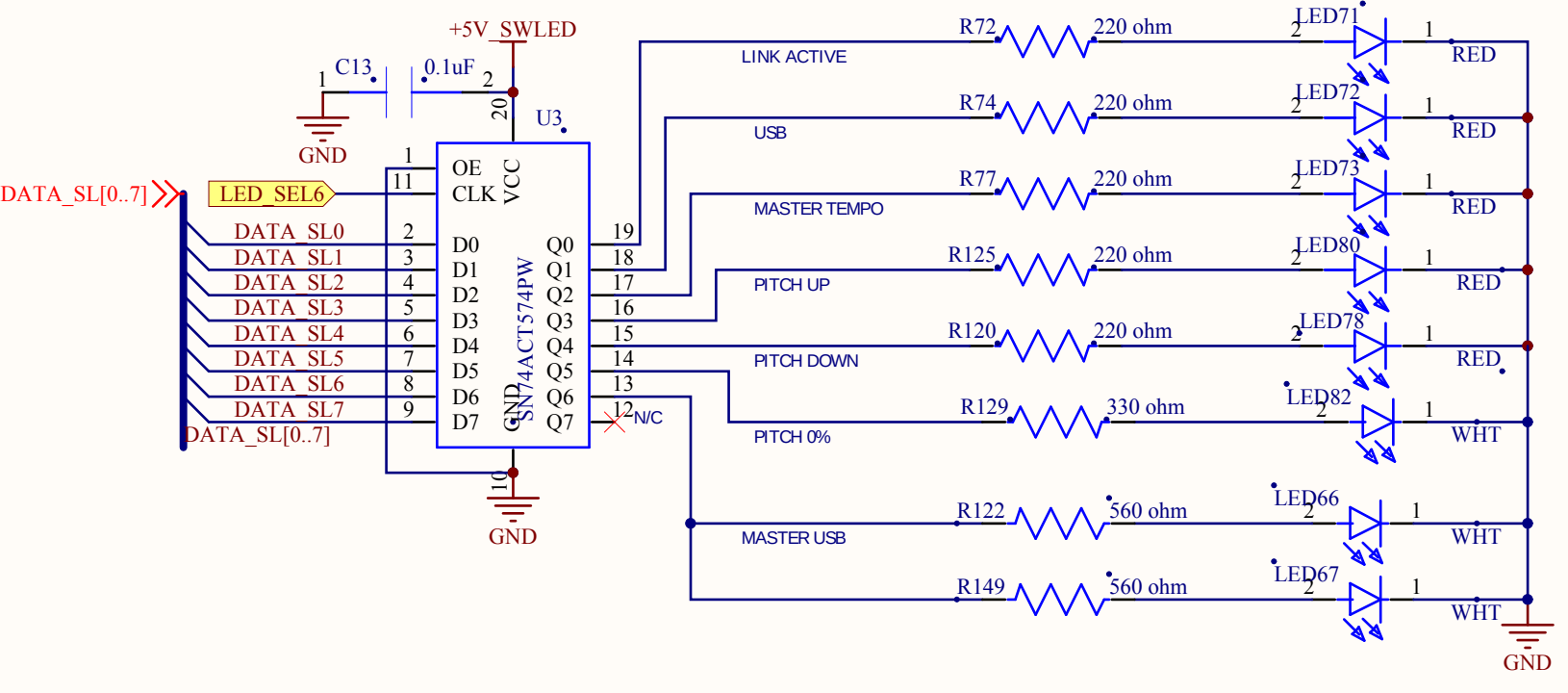
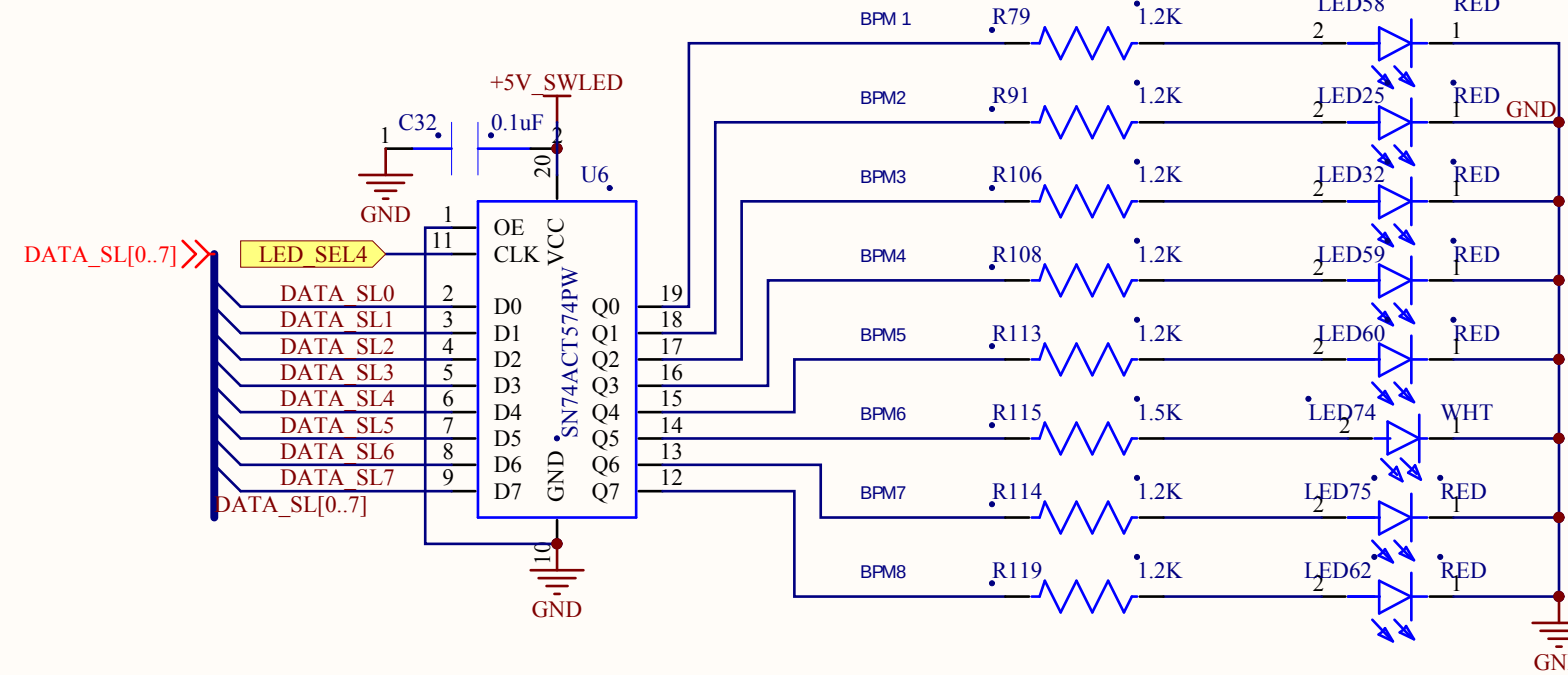
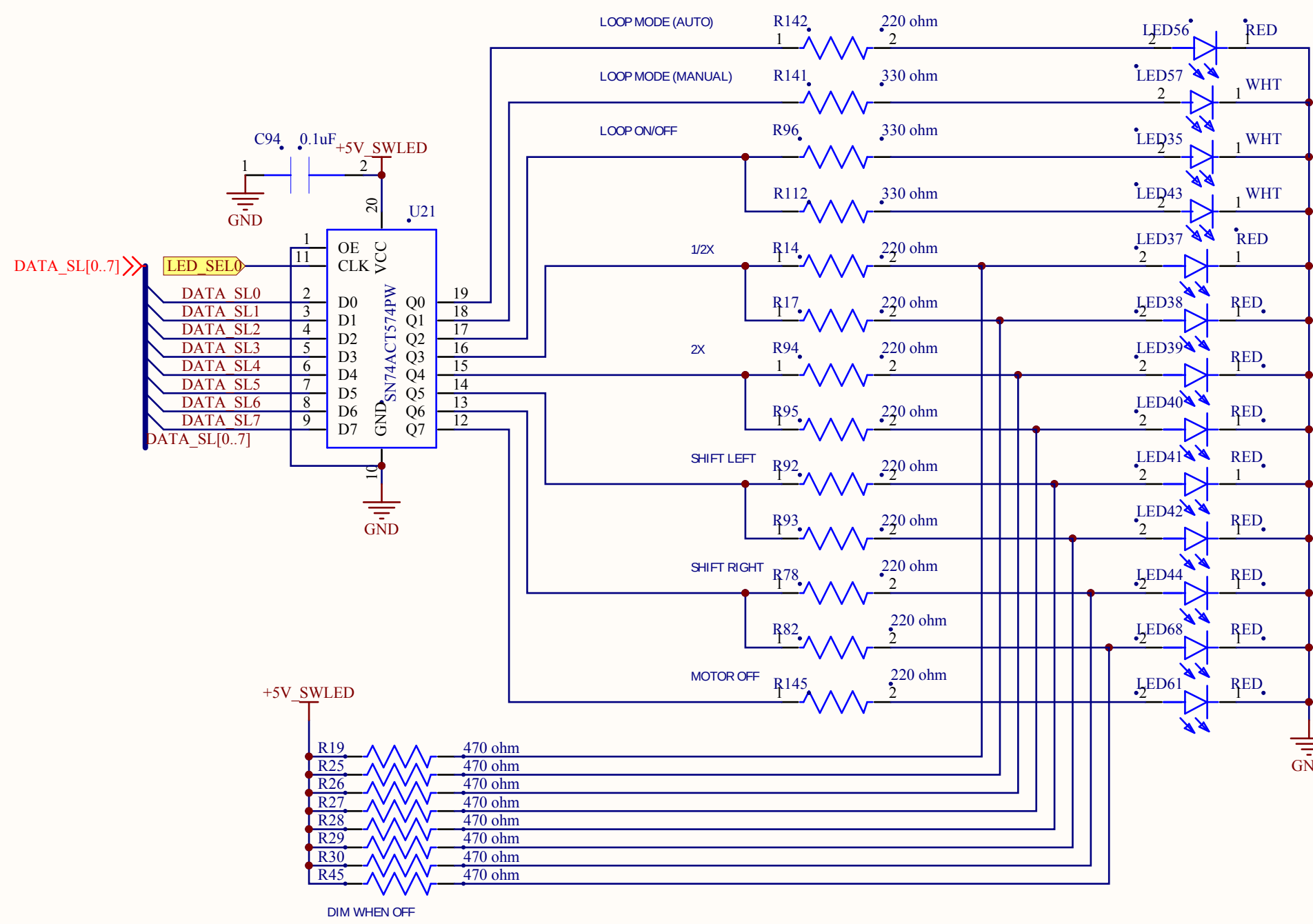






DISCARD



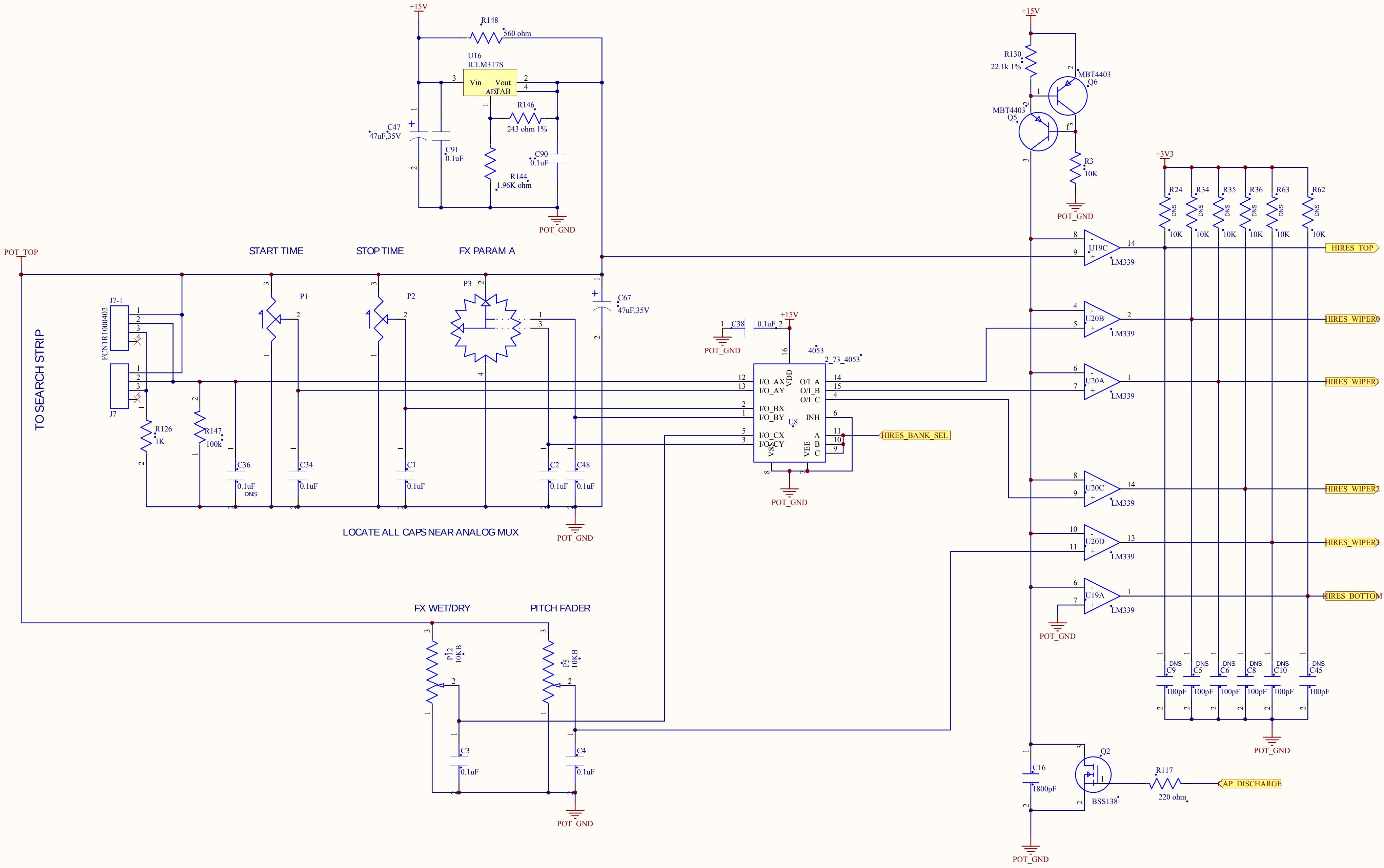


|                                                                                           |  |       |  |        |                   |     |        |             |  |          |
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|  GND |  |       |  |        | Title:            |     |        |             |  |          |
|                                                                                           |  |       |  |        | Size              |     | Number |             |  | Revision |
|                                                                                           |  |       |  |        | A2                |     |        |             |  |          |
|                                                                                           |  |       |  |        | Date : 06/11/2009 |     |        | Sheet of    |  |          |
|                                                                                           |  |       |  |        | File Name:        |     |        | Design By : |  |          |
| Approval                                                                                  |  | Check |  | Design |                   | REV | DATE   | Description |  |          |

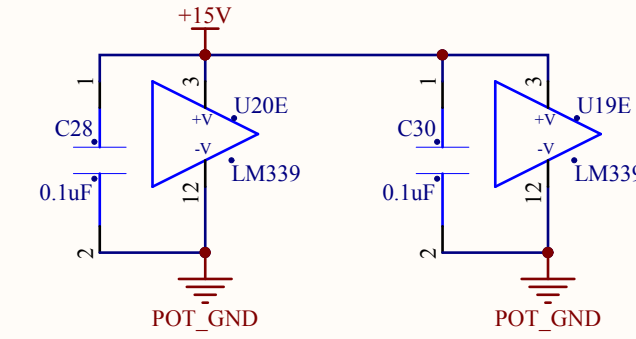




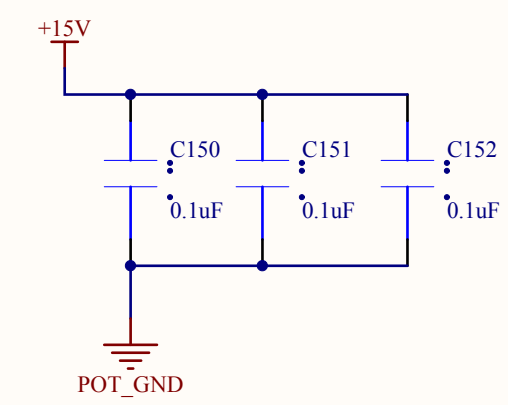
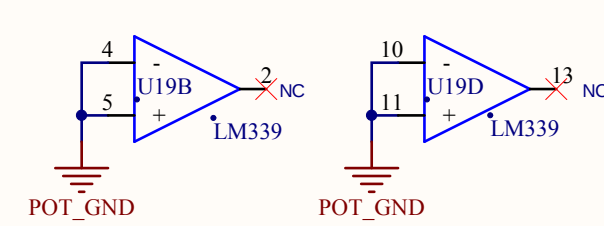




DECOUPLING

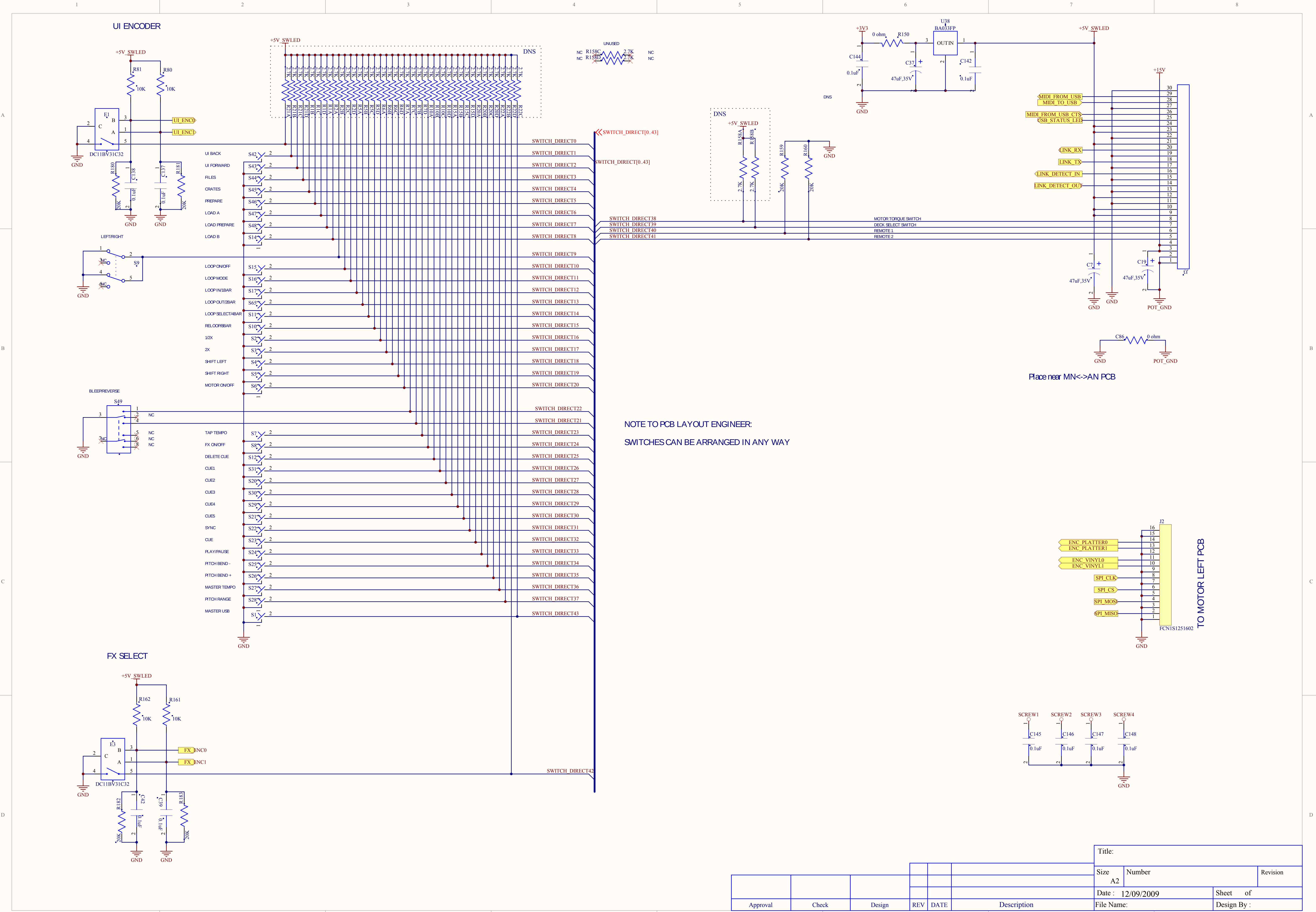


UNUSED



|          |       |        |     |      |             |                   |        |          |
|----------|-------|--------|-----|------|-------------|-------------------|--------|----------|
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|          |       |        |     |      |             | Size              | Number | Revision |
|          |       |        |     |      |             | A2                |        |          |
|          |       |        |     |      |             | Date : 06/02/2009 |        |          |
|          |       |        |     |      |             | Sheet of          |        |          |
|          |       |        |     |      |             | File Name:        |        |          |
|          |       |        |     |      |             | Design By :       |        |          |
| Approval | Check | Design | REV | DATE | Description |                   |        |          |





NOTE TO PCB LAYOUT ENGINEER:

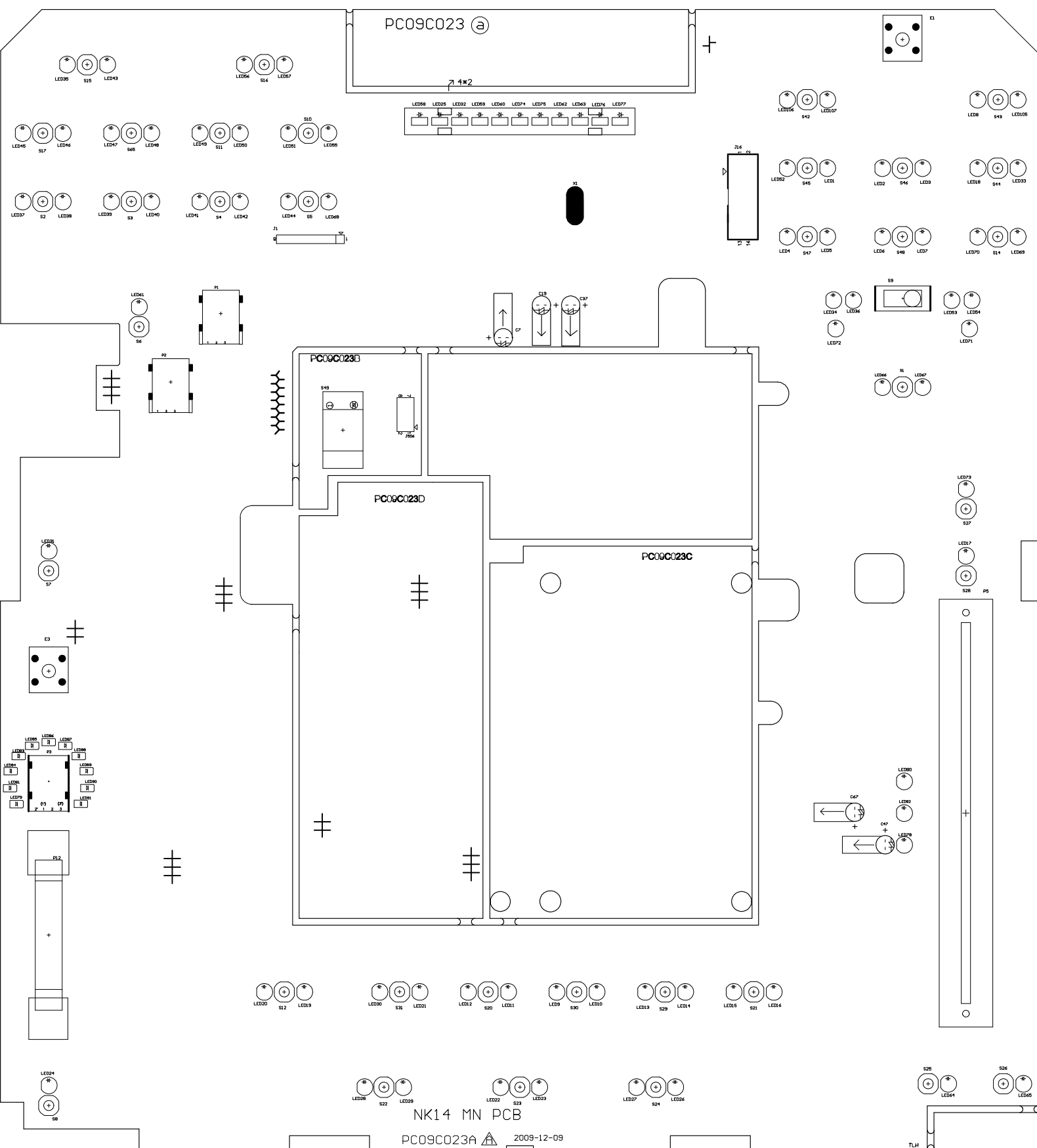
SWITCHES CAN BE ARRANGED IN ANY WAY

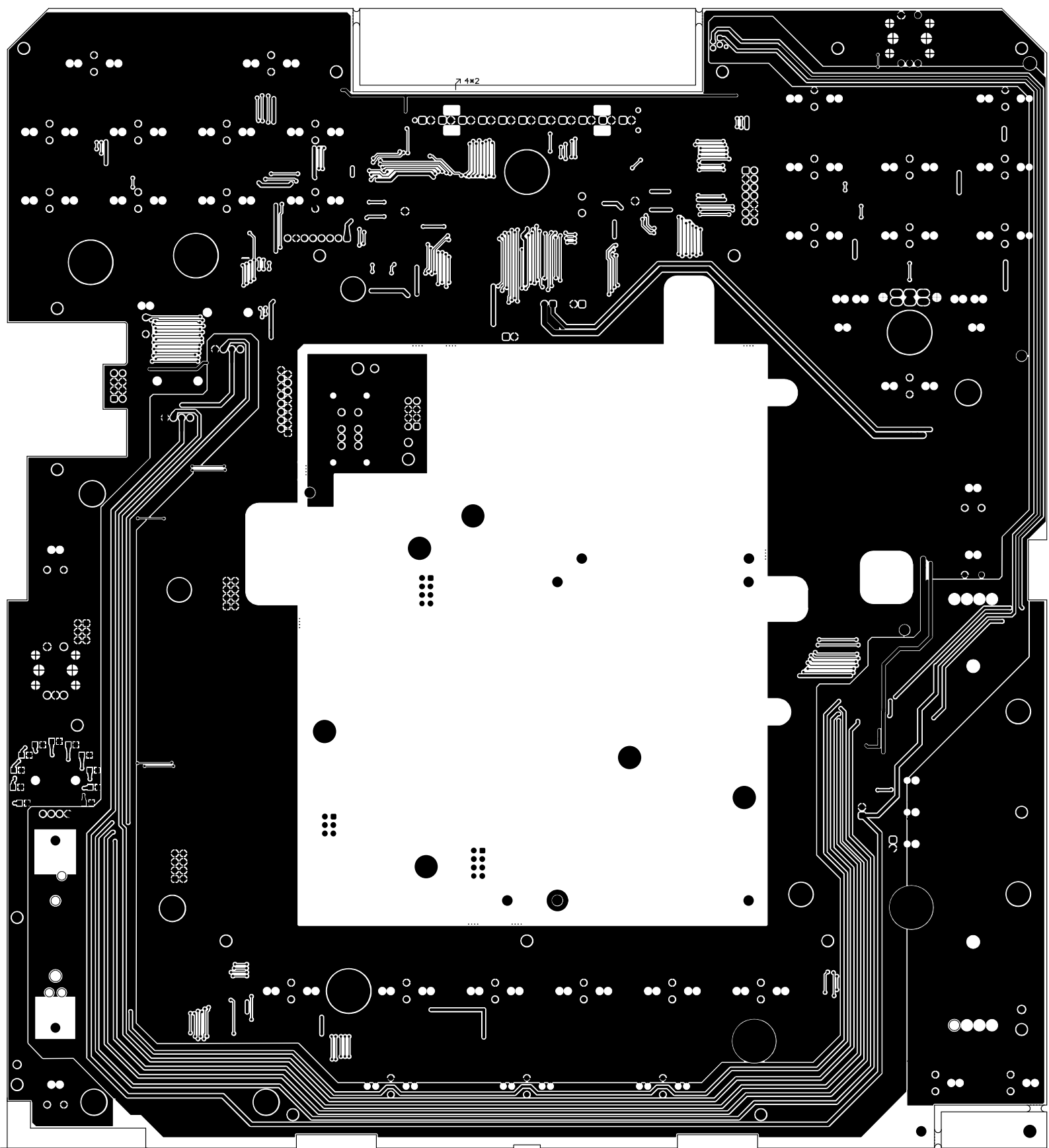
Place near MN<->AN PCB

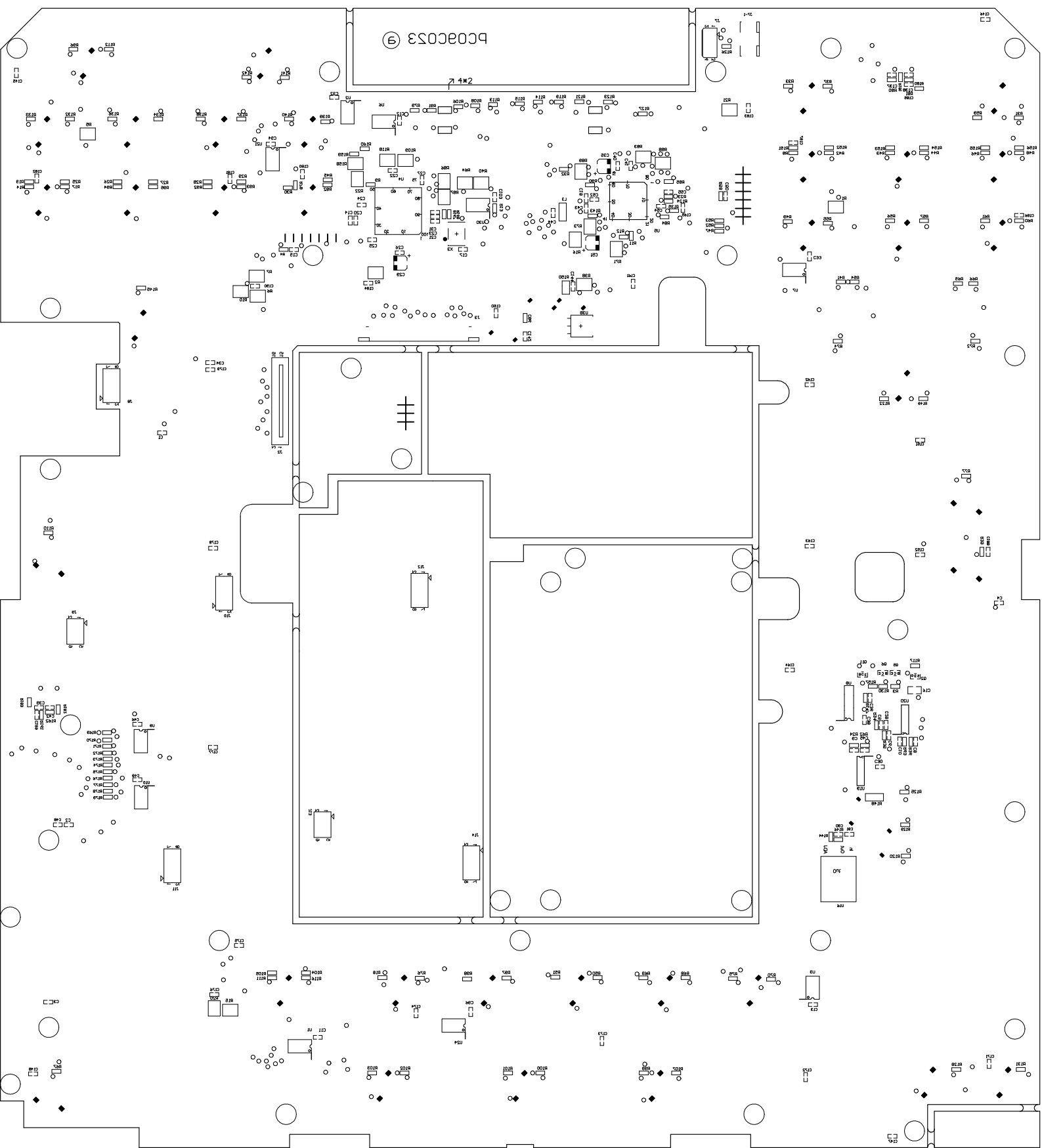
TO MOTOR LEFT PCB

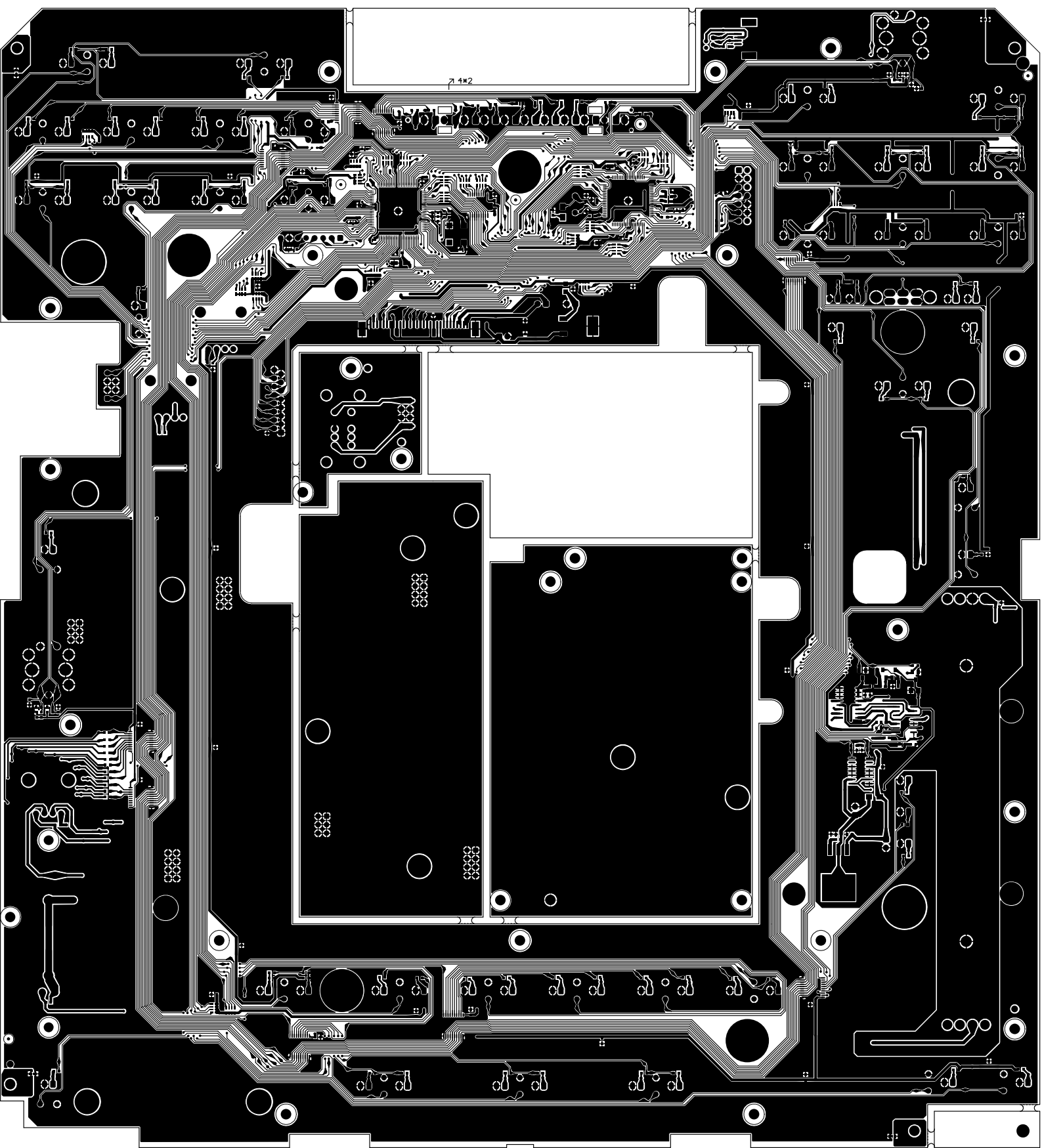
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|          |       |        |     |      | Size<br>A2        | Number |             | Revision |
|          |       |        |     |      | Date : 12/09/2009 |        |             |          |
| Approval | Check | Design | REV | DATE | Description       |        | File Name:  | Sheet of |
|          |       |        |     |      |                   |        | Design By : |          |



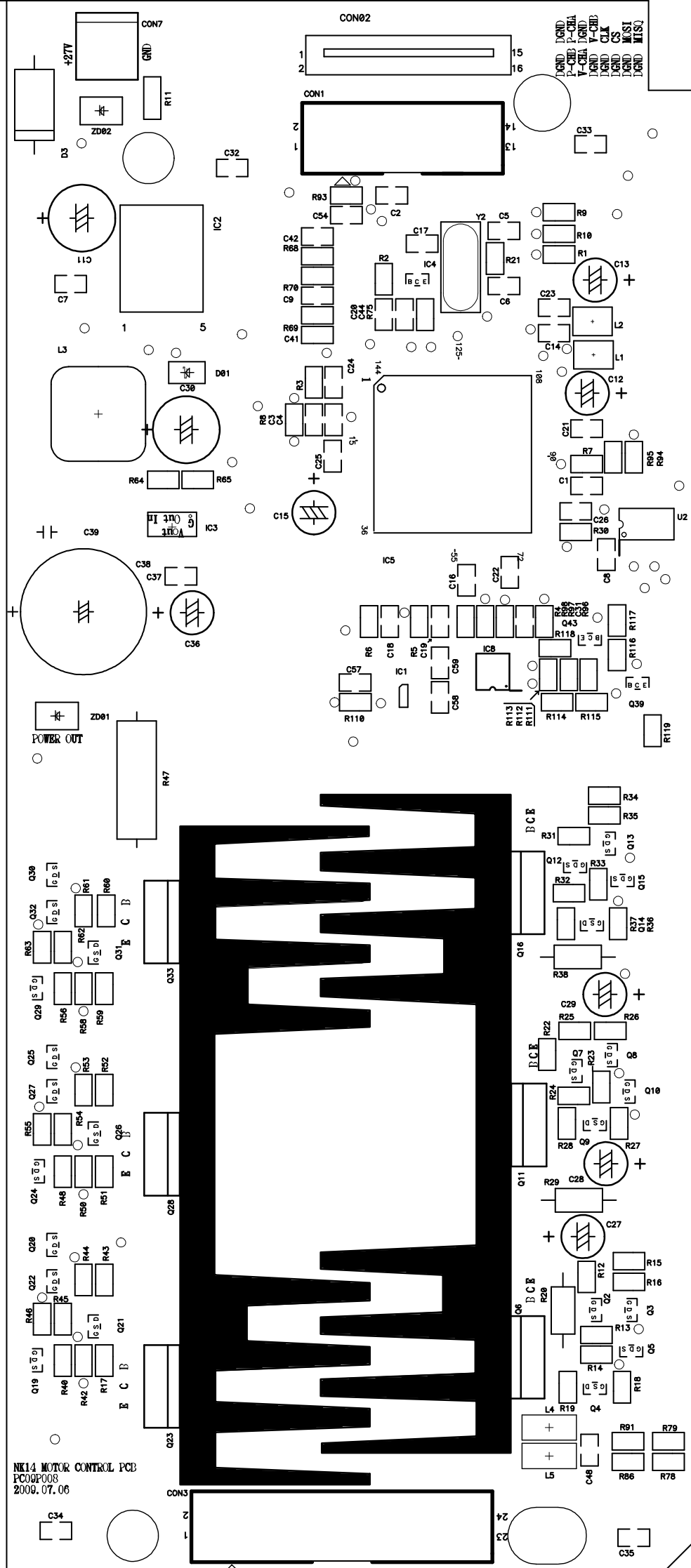












NE14 MOTOR CONTROL PCB  
PC09P008  
2009.07.06

